Mastermind Revision Notes

GRADE 8 · SCIENCE

Fast, high-retention revision notes built directly from the NCERT / RBSE Grade 8 Science textbook.

PART 1 — CHAPTERS 1 to 7

How to use these notes:

- Each chapter follows the textbook's own section order — read straight through.
- Every **bold term** is a keyword you must be able to define or identify.
- Tables compare and classify — memorise the rows, they are common exam questions.
- Blue **Note** boxes hold formal definitions and formulas.
- Gold **Key Takeaways** at the end of each chapter = your last-2-minutes revision.

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CHAPTER 01

Exploring the Investigative World of Science

Grade 8 · Science

This chapter is an **invitation to think like a scientist**. It shows that science is not just a body of facts but a **method** — a way of asking focused questions, designing simple experiments, and using observations to build understanding. The textbook's running example is the everyday puzzle of why a **puri puffs up** and why one side of it is thinner.

1.1 What Investigation in Science Means

- **Science begins with wonder** — simple "Why?" and "How?" questions about the world around us.
- In Grade 6 you learnt that science begins with wonder; in Grade 7 that **science is always evolving** — each answer opens new questions.
- In Grade 8 you enter the **Investigative World**: you learn not just new facts but **how to find new facts**.
- Investigation means asking **focused questions**, designing **simple experiments**, and using observations to **improve understanding**.
- You do not need a fancy laboratory — the **kitchen at home** is a wonderful place to observe and ask questions.

1.2 The Scientific Method — Using the Puri Example

The textbook walks through a full mini-investigation of why a puri puffs up. This is a model for **any** scientific investigation.

Step 1 — Ask a scientific question

What different things may change the way a puri puffs up when fried?

Step 2 — Identify what you can change (variables) and what you can observe

What we can change (control)	What we can observe / measure
Thickness and size of rolled dough	Does the puri puff up? (yes / no)
Type of flour (atta, maida, etc.)	Time taken to puff up (seconds)
Temperature of the hot oil	Does a very thick layer still give a thin side?
How the dough is dropped into oil (vertical, at an angle, slowly)	Oil splatter, smell, smoke

Step 3 — Change only one thing at a time

- If testing the effect of oil temperature, use circles of dough of the **same thickness** and drop them in the **same way**.
- This is called a **controlled experiment** — it is how all scientific experiments, simple or complex, are done.

Step 4 — Keep notes and ask more questions

- Record everything you see and sense: Did the oil splatter, smell, or smoke?
- New questions follow: Do puris puff better when made fresh or from stored dough? What happens if I prick a hole before frying?

> **IMPORTANT**
> Even this simple everyday observation — of a puri puffing — is **not completely understood by scientists today**. Science does not always have final answers.

1.3 Symbols Used in This Textbook

Symbol	Meaning

Root (bottom of left-hand pages)	Deep, solid foundation of knowledge — stay grounded in real observations, traditions, and heritage.
Kite (top corner of right-hand pages)	Curiosity must take flight — allow ideas to soar towards new horizons.
Patterns in bottom lines	Hidden scientific thoughts for interest.

Balance: investigation works best when we balance the **solid ground of careful observation** with the **freedom of creative thinking**.

★ KEY TAKEAWAYS

- Science is a **method**, not just a set of facts — ask focused questions, experiment thoughtfully, explain clearly.
- Change **one variable at a time** while keeping other conditions the same.
- Keep careful notes of everything you see and sense during an experiment.
- Every answer opens new questions — science is always evolving.
- Simple everyday phenomena (like a puri puffing) can be genuine scientific investigations.

CHAPTER 02

The Invisible Living World: Beyond Our Naked Eye

Grade 8 · Science

PROBE & PONDER

- What might you see if the invisible world around you became visible?
- How might this hidden world change how you think about size, complexity, or what counts as 'living'?
- How do these tiny living beings interact with each other?

This chapter opens up the world too small for the naked eye. It begins with the invention of the **lens** and **microscope**, then builds up from the **cell** — the basic unit of life — to **tissues, organs, organ systems**, and finally to **microorganisms** and their many roles in our lives.

2.1 The Invention of the Microscope

- Long ago, people discovered that a **curved piece of glass** could make small things look bigger.
- The glass was shaped like a **lentil seed** — thick in the middle, thin at the edge — hence called a **lens**.
- Over time, lenses improved; each new tool (magnifying glass → microscope) helped humans see what their eyes could not.
- The **microscope** opened a hidden world of tiny living creatures.
- A **round-bottom flask filled with water** acts like a magnifying glass (Fig. 2.1).
- Scientists have also created a **low-cost, foldable paper microscope** — it may not give the detail of high-powered lab microscopes, but it makes the microscopic world accessible to many people.

2.2 What Is a Cell?

DEFINITION

A **cell** is the basic structural and functional unit of all living organisms. All living beings are made up of cells.

2.2.1 Basic parts of a cell (from the onion peel & cheek cell activities)

Part	Description	Function
Cell membrane	Outer layer; thin membrane	Encloses cytoplasm and nucleus; separates one cell from another; porous — allows entry of materials needed for life and exit of waste
Cytoplasm	Jelly-like material between membrane and nucleus	Contains other cell components and compounds (carbohydrates, proteins, fats, mineral salts); most life processes occur here
Nucleus	Round structure in the middle, covered by a thin membrane	Regulates all activities within the cell; regulates growth
Cell wall	Extra outer layer — present in plant cells (e.g. onion peel)	Provides rigidity and strength to plants; cells arranged compactly and look firm

2.2.2 Additional structures (A Step Further)

- **Plastids** — tiny rod-shaped structures in plant cells. **Chloroplasts** contain **chlorophyll** (green), help in photosynthesis; in non-green parts they help store substances.
- **Vacuole** — large, empty-looking space in plant cells. Helps store important substances, get rid of waste, and maintain cell shape (gives strength and support to the plant).
- Animal cells: vacuoles usually **not present**; if present, they are **small** and store certain dissolved substances.

2.2.3 Plant vs. animal cell — comparison

Feature	Plant cell	Animal cell

Cell wall	Present	Absent
Chloroplast	Present (green parts)	Absent
Vacuole	Large, central	Usually absent; if present, small
Shape	Rigid, often rectangular	Irregular, often rounded

2.3 Variation in Shape and Structure of Cells

The unique shape, size, and structure of cells help them carry out their **specific functions**.

Cell type	Shape / structure	Function
Cheek cells	Thin and flat, polygon-shaped	Form a protective lining on the inner surface of the cheek
Muscle cell	Spindle-shaped, thin, flexible	Contract and relax in a wave-like manner (e.g. push food down the food pipe)
Nerve cell (neuron)	Very long with branches	Carry messages in the body; elongated branched shape reaches different parts and passes messages quickly
Plant cells	Rectangular, elongated, oval, or tube-like	Some form long tubes that carry water throughout the plant
Stomach cells	Different types	Muscle cells churn food; inner lining cells produce digestive juices and acid

2.4 Levels of Organisation in the Body

> **THE ORGANISATION LADDER**
>
> **Cell → Tissue → Organ → Organ system → Organism**

- **Cell** — basic unit of life (like a brick is the basic unit of a wall).
- **Tissue** — a group of similar cells.
- **Organ** — different tissues organised together (e.g. stomach).
- **Organ system** — several organs working together to perform a major function (e.g. digestive system).
- **Organism** — all organ systems together make a complete living being.
- The life of complex organisms begins with a **single cell — the egg**.
- The egg divides repeatedly to form a complete living being made of many cells → **multicellular organisms** (animals including humans, and plants).

> **EVER HEARD OF…**
>
> The **yolk of an ostrich egg** is a single cell — the **largest known cell** in the living world — measuring about **130–170 mm** in diameter. The egg contains extra non-cellular material: a shell for protection and a white liquid that nourishes the cell.

2.5 What Are Microorganisms?

> **DEFINITION**
>
> **Microorganisms (microbes)** are living organisms too small to be seen with the naked eye. (micro = very small; organisms = living beings)

- Found everywhere — in water, soil, air, and even inside our body.
- Some are **unicellular** (single-celled): bacteria, Amoeba, Paramecium, yeast.
- Some are **multicellular**: some fungi, some algae, moulds.
- They vary in shape — spherical, rod-shaped, comma-shaped, spiral, or irregular.

- Some live in **extreme conditions** — hot water springs, snow-cold zones — as well as moderate temperatures.

2.5.1 Microorganisms observed in pond water (Table 2.1)

Microorganism	Group	Recorded data
Amoeba	Protozoa	Single cell, moving, irregular shape
Paramecium	Protozoa	Single cell, moves from place to place with specialised structures
Algae	Algae	Single cell, green pigment; movement with specialised structures

2.5.2 Microorganisms observed in soil suspension (Table 2.2)

Microorganism	Group	Remarks
Bread mould	Fungi	Branched filament without chlorophyll; sac-like structure
Mould	Fungi	Branched filament without chlorophyll; brush-like structure
Algae	Algae	Spherical; presence of chlorophyll (green pigment)
Bacteria	Bacteria	Spherical, comma, spiral or rod-shaped; one long hair-like structure and many small hair-like projections

> **EVER HEARD OF...**
> **Viruses** are microscopic and **acellular**. They multiply only when they enter a living cell. They may infect plants, animals, or bacterial cells and cause disease.

2.6 How Are We Connected to Microbes?

Microbes are found everywhere — water, soil, air, food. This is why food left outside rots with a powdery or cotton-like growth.

- **Why pickles and murabbas don't get infected:** They contain many spices with **salt or sugar** which act as **preservatives**. High concentration of salt or sugar does not allow microbes to grow.
- Some microbes live in our **gut** and help in **digestion**.

2.6.1 Key players in cleaning the environment

- Fungi and bacteria act on plant waste and slowly break it down into simpler, **nutrient-rich manure**.
- **Decomposition** — the breakdown of complex substances of fallen leaves and dead animals into simpler nutrient-rich substances.
- Nutrients return to the soil and help plants grow better.
- Manure formation occurs at **optimal temperature** and **appropriate moisture**.

> **A STEP FURTHER — BIOGAS**
> Some bacteria and fungi live in an **oxygen-free environment**. They decompose plant and animal waste (or household wastewater) and release a mixture of gases called **biogas** — mainly carbon dioxide and a high proportion of **methane**. Used as fuel for cooking, heating, generating electricity, and running vehicles.

> **BE A SCIENTIST — ANANDA MOHAN CHAKRABARTY (1938–2020)**
> Developed a special bacterium (1971) that could **break down oil spills**. Received a patent in 1980. (Patent = a copyright given to a person so no one else can copy, use, or sell their invention without permission.)

2.6.2 Microorganisms and food

Microbe	Process	Product / use
Yeast (a fungus)	Respires and breaks down food; releases **carbon dioxide** (bubbles make dough soft and fluffy) and a small amount of **alcohol** (gives different smell)	Breads, cakes, pastries, idli/dosa/bhatura batter (with bacteria)

Lactobacillus (bacterium)	Feeds on lactose (milk sugar), multiplies, ferments milk → produces **lactic acid** (makes curd sour)	Curd formation; fermentation of batter for idli, dosa, bhatura
Rhizobium (bacterium)	Lives in root nodules of legumes; **traps nitrogen** from air and makes it useful for plants	Natural fertiliser; farmers grow legumes in rotation with other crops

> **IMPORTANT**
> **Yeast** grows well in **warm** conditions (hence sugar + warm water). **Lactobacillus** also grows well in **warm** conditions — curd forms in warm places but not in a refrigerator.

2.6.3 Microalgae — tiny helpers in water

- Microscopic plant-like organisms living in water, soil, air, and even on trees.
- Make their own food using sunlight → release **oxygen**; produce **more than half of Earth's oxygen supply**.
- Rich in nutrients; food source for many aquatic animals.
- Examples: **Spirulina, Chlorella, Diatoms** — used as health supplements and medicines.
- Help clean water and are used to make **biofuel**.
- Threatened by pollution, climate change, and habitat destruction.

> **EVER HEARD OF... SPIRULINA**
> A microalga called a **superfood** because of its health benefits. A good source of **vitamin B_{12}**; rich in **protein — more than 60% of its body weight** — with only a small amount of fat and sugar. Can be farmed easily (clear glass tank, bright place away from direct sunlight, pond water, stir twice a week, harvest after 3–6 weeks).

2.7 Why Is the Cell Considered the Basic Unit of Life?

- The body of all living organisms is made of cells.
- A single cell contains various components that help organisms perform functions.
- **Multicellular** organisms — many cells; cells carry out specialised functions individually but also **cooperate** to increase chances of survival.
- **Unicellular** organisms — a single cell carries out all functions necessary for survival.
- **Yeast** = unicellular fungus; **mould** = multicellular fungus.

2.7.1 Cell features that distinguish groups

Feature	Bacteria	Fungi	Plant / animal cells
Cell membrane	Present	Present	Present
Cell wall	Present	Present	Present (plants); absent (animals)
Chloroplast	Absent	Absent	Present in plant green parts
Nucleus	**Nucleoid** (no well-defined nucleus / nuclear membrane)	Well-defined nucleus	Well-defined nucleus

- **Electron microscope** — magnifies the cell about **10,00,000 times**; used to see subcellular components.

★ KEY TAKEAWAYS

- Cell = basic unit of life. Three main parts: **cell membrane, cytoplasm, nucleus**; plant cells also have a **cell wall**.
- Levels of organisation: **Cell** → **Tissue** → **Organ** → **Organ system** → **Organism**.
- Cell shape is related to function (cheek = flat protective; nerve = long branched; muscle = spindle, flexible).
- Microorganisms are everywhere — unicellular (bacteria, protozoa, yeast) or multicellular (moulds, some algae).
- **Yeast** → CO_2 + alcohol → bread/cakes. **Lactobacillus** → lactic acid → curd. **Rhizobium** → nitrogen fixation in legume root nodules.
- **Decomposers** recycle waste into nutrient-rich manure; some bacteria produce **biogas**.
- **Viruses** are acellular and multiply only inside a living cell.
- Bacteria lack a well-defined nucleus — they have a **nucleoid**.

CHAPTER 03

Health: The Ultimate Treasure

Grade 8 · Science

PROBE & PONDER

- How does your body respond to an infection such as the common cold?
- We rarely see smallpox or polio these days, but diabetes and heart problems are more common. Why?
- Could climate change lead to new types of diseases?
- How do emotions like stress or worry affect us and make us sick?
- Why do some groups of people get affected more than others during disease outbreaks?

This chapter defines health as far more than the absence of disease. It covers **communicable and non-communicable diseases**, how diseases spread, prevention (including vaccines and immunity), treatment (antibiotics and antibiotic resistance), and the role of lifestyle and environment in staying healthy.

3.1 Health: Is It More Than Not Falling Sick?

WHO DEFINITION

Health is a **'state of complete physical, mental, and social wellbeing, and not merely the absence of disease'**.

- A healthy person can generally perform tasks more efficiently and cope well in different and difficult situations.
- A healthy person can adjust well with peer groups and other members of society.
- Three aspects of health: **Physical, Mental, Social**.
- Health also includes feeling good physically, staying positive, and having strong relationships.

OUR SCIENTIFIC HERITAGE — AYURVEDA

True health is a **balance of body, mind, and surroundings**. Following **dinacharya** (daily routine) and **ritucharya** (seasonal routine) helps maintain this balance. Eating fresh, wholesome food suited to one's **prakriti** (body constitution) is essential. Regular exercise, cleanliness, restful sleep, and a calm mind support overall wellbeing — through yoga, meditation, and mindfulness.

3.2 How Can We Stay Healthy?

We must eat nutritious food, maintain hygiene, stay in a clean place, exercise regularly, get proper sleep, spend time with family and friends, and have a positive attitude.

3.2.1 Maintain a healthy lifestyle

- Eat a **balanced diet** — plenty of fruits, vegetables, whole grains. Avoid processed, fatty, or sugary food and drinks.
- Stay **physically active** — playing outdoors, walking, running, cycling, exercising.
- Limit screen time; spend more time in nature.
- Get enough **sleep** — helps body and mind rest and recover.
- Practice **yoga or pranayama** regularly.
- Say **'NO'** to tobacco, alcohol, and addictive drugs.

3.2.2 Keep the environment clean

- Clean air and water are important for health. In cities, air pollution from vehicles and factories can cause coughing or asthma.
- **Air Quality Index (AQI)** — helps us know how clean the air is.
- Spending time with friends and family, talking, laughing, and having fun help keep our minds healthy too.

3.3 How Do We Know That We Are Unwell?

Term	Meaning	Example
Symptom	What we **feel**	Pain, tiredness, dizziness
Sign	Something that can be **seen or measured**	Fever, rash, high blood pressure, swelling

3.4 Diseases: Causes and Types

> **DEFINITION**
> A **disease** is a condition that affects the normal working of the body or mind. It can happen when one or more organs or organ systems stop functioning properly.

- **Pathogens** — disease-causing organisms: bacteria, viruses, fungi, worms, protozoa (single-celled organisms).
- Other diseases result from **poor nutrition** or an **unhealthy lifestyle**.

Type	Cause	Spread?	Examples
Communicable diseases	Pathogens	Yes — from one person to another	Typhoid, dengue, flu, chickenpox, COVID-19
Non-communicable diseases (NCDs)	Lifestyle, diet, and/or environment (not pathogens)	No	Cancer, diabetes, asthma; deficiency diseases (scurvy, anaemia, goitre)

- In recent years, NCDs like diabetes, heart disease, and cancer have become more common in India — due to changes in how people live: more processed food, less exercise, longer lives.
- Today, **most deaths in India are caused by NCDs**.
- Diseases that persist for a long time (more than 3 months) are called **chronic diseases** (e.g. cancer, diabetes, asthma).

3.4.1 How communicable diseases spread (Fig. 3.4)

Method	How
Direct contact	Shaking hands
Indirect contact	Sharing personal items of an infected person
By air	Infected person coughs or sneezes
By food / water	Contaminated drinking water or food
Vectors (insects)	Mosquitoes, houseflies carry pathogens
Rabid animals	Bites from infected animals

3.4.2 Common communicable diseases (Table 3.1)

Disease	Causal agent	Site of infection	Key symptoms	Preventive measures
Common cold & influenza	Virus	Respiratory tract	Nasal congestion/discharge, sore throat, fever, cough, body ache	Wash hands frequently, don't share personal items, cover mouth and nose with a mask
Chickenpox	Virus	Respiratory tract, skin	Mild fever, itchy skin, rashes, blisters	Complete isolation, cover mouth and nose, vaccination
Measles	Virus	Skin, respiratory tract	Fever, sore throat, reddish rashes on neck, ears and other parts	Isolation, cover mouth and nose, good hygiene, vaccination
Tuberculosis (TB)	Bacteria	Lungs	Cough, fever, fatigue, loss of appetite, night sweats	Avoid close contact with TB-infected people, cover mouth

				and nose, good hygiene, vaccination
Hepatitis A	Virus	Liver	Fatigue, fever, loss of appetite, nausea, vomiting, jaundice, pain in upper right abdomen	Drink boiled water, vaccination
Cholera	Bacteria	Intestine	Diarrhoea and dehydration	Personal hygiene, good sanitation, properly cooked food and boiled water, vaccination
Typhoid	Bacteria	Intestine	Headache, abdominal discomfort, fever, diarrhoea	Personal hygiene, good sanitation, cooked food and boiled water, vaccination
Ascariasis (roundworms)	Worms	Intestine	Worms in stool, loss of appetite, poor growth, diarrhoea, weight loss, anaemia	Personal hygiene, good sanitation, properly cooked food and boiled water
Malaria	Protozoa	Skin, blood	High fever, profuse sweating, periodic chills	Mosquito nets/repellents, long-sleeved clothes, control mosquito breeding
Dengue fever (Break bone fever)	Virus	Skin, blood	Fever, headache, muscle and joint pain, nausea	Mosquito nets/repellents, long-sleeved clothes, control breeding, avoid still water

3.4.3 Simple precautions against communicable diseases

- Keep yourself and your surroundings clean.
- Practise basic hygiene every day.
- Wash hands with soap and water to remove pathogens.
- Cover your mouth and nose while coughing or sneezing.
- Wear a mask in crowded places.
- Avoid sharing personal items like towels and handkerchiefs.
- Keep home, food, and water clean.
- Stay home and rest when unwell — helps the body recover and minimises spreading the disease.

3.4.4 How non-communicable diseases are caused

- Linked to **lifestyle, diet, and/or environment**; most common cause of death in India.
- **Deficiency diseases** (scurvy, anaemia, goitre) are also non-communicable — caused by lack of specific nutrients.
- **Diabetes** — common; becoming more prevalent in adults and children. India has one of the highest numbers of people with diabetes in the world. Develops due to a combination of hormonal imbalances, unhealthy eating habits, lack of physical activity, being overweight or obese, and other reasons.

> **BE A SCIENTIST — KAMAL RANADIVE (1917–2001)**
> Pioneering biomedical researcher. Studied how **hormones and certain viruses are linked to cancer**, helping improve its treatment and prevention. Her work also showed how **tobacco, diet, and pollution** can raise the risk of cancer, highlighting the importance of a healthy lifestyle.

3.5 How to Prevent and Control Diseases?

'Prevention is better than cure.'

3.5.1 Immunity and vaccines

> **DEFINITIONS**
> **Immunity** — the natural ability of our body to fight diseases. The body has a special **immune system** that helps fight against diseases.

Vaccine — helps our body fight certain diseases by **training the immune system** to recognise and attack harmful germs.

Acquired immunity — protection developed after exposure to a pathogen or a vaccine.

- Vaccines can be made from **weakened or dead pathogens**, or from **inactive or harmless parts** of the pathogen.
- Some newer vaccines instruct our own body cells to make a harmless part of the germ, which the immune system then learns to fight.
- Example: **tetanus shot** — contains an inactivated bacterial toxin that helps the immune system develop protection without causing disease.
- Vaccines are **preventive, not curative** — they minimise serious diseases before they happen but do not treat them once someone is already sick.
- Getting vaccinated protects you **and** the people around you.

3.5.2 Edward Jenner and the smallpox vaccine

Step	Detail
Observation	Milkmaids who had **cowpox** did not catch **smallpox** (the two viruses are related)
Hypothesis	Content in the pus of cowpox blisters protected people from smallpox
Experimentation	Injected cowpox sap into a boy, who later showed no illness when exposed to smallpox
Results	People infected with cowpox sap were now resistant to smallpox
Application	Mass vaccination eventually helped **eradicate smallpox worldwide**

OUR SCIENTIFIC HERITAGE — VARIOLATION

Long before modern vaccines, India had a traditional method called **variolation** to protect against smallpox. It involved using material from a smallpox sore to scratch the skin and create a mild infection to build immunity. People who performed this practice were known as **teekedaars**.

EVER HEARD OF... INDIA'S ROLE IN VACCINE PRODUCTION

India is one of the world's **largest vaccine producers**. It manufactures vaccines on a massive scale and supplies them to many countries. Indian vaccine companies played a key role during the **COVID-19 pandemic** and continue to support global health efforts.

BE A SCIENTIST — MAHARAJ KISHAN BHAN

Well-known Indian doctor and scientist. As Secretary of the Department of Biotechnology, he helped promote science and innovation in India. Played a key role in developing the **Rotavirus vaccine**, which protects children from diarrhoea. He believed in using research to create affordable healthcare.

3.5.3 Treatment of diseases

- If our immune system fails, we fall ill and need a doctor.
- **Antibiotics** — medicines that kill **bacteria**. They work only against bacterial infections because they target parts of bacterial cells that are different from human or other animal cells.
- Antibiotics **do not work** against viruses or diseases caused by protozoa.

DISCOVERY OF PENICILLIN — ALEXANDER FLEMING (1928)

While studying harmful bacteria, Fleming noticed that a **mould** on a discarded petri dish stopped the bacteria from growing. He realised the mould released a substance that killed the bacteria. This chance discovery led to **penicillin**, the first antibiotic used to treat bacterial infections.

3.5.4 Antibiotic resistance (Fig. 3.5a & 3.5b)

DEFINITION

Antibiotic resistance — a phenomenon where bacteria that were earlier killed by a given antibiotic survive and multiply despite treatment with that antibiotic.

- Caused by **indiscriminate use** of antibiotics — taking them when not required, or taking an incomplete dose.
- Resistance can develop in bacteria in a human body and in **animals** (when antibiotics are given to cattle).
- Spread through **animal food products**, **soil via manure/animal faeces**, **crop produce**, and **humans** (including in hospitals through dirty surfaces and healthcare workers).
- Makes common infections **harder to treat** and increases the risk of complications, prolonged illness, and even death.

How to reduce antibiotic resistance:

- Use antibiotics **wisely** — only when prescribed by a doctor.
- Take the **correct dose** and for the **right duration**.
- Avoid unnecessary use — helps prevent the rise of resistant bacteria and keeps antibiotics effective for future generations.
- Do not give antibiotics to cattle without veterinary advice.

3.5.5 Traditional medicine systems

- **Ayurveda, Siddha, Unani** — used in India for many years to manage common health problems.
- Use natural substances like herbs, oils, and minerals; focus on a healthy lifestyle and balanced diet.
- Can help with some conditions and everyday wellbeing, but **may not be effective for all diseases and at all stages**.

3.5.6 Treatment of non-communicable diseases

- Focus on **managing symptoms** and improving quality of life through medication, lifestyle changes, and rehabilitation.
- **Early diagnosis** and **continuous care** are key to control disease progression and prevent complications.

★ KEY TAKEAWAYS

- WHO health = **physical + mental + social wellbeing**, not merely absence of disease.
- **Symptom** = what we feel; **Sign** = what can be seen or measured.
- **Communicable** diseases spread from person to person (pathogens); **non-communicable** do not (lifestyle/diet/environment).
- Diseases spread by air, direct/indirect contact, food/water, vectors, and rabid animals.
- **Vaccines** are preventive, not curative — they train the immune system.
- **Antibiotics** work only on bacteria, not viruses. Misuse causes **antibiotic resistance**.
- India is one of the world's largest vaccine producers.
- NCDs (diabetes, heart disease, cancer) now cause most deaths in India — prevention through healthy lifestyle is key.

CHAPTER 04

Electricity: Magnetic and Heating Effects

Grade 8 · Science

PROBE & PONDER

- If we don't have an electric lamp while making a circuit with a cell, is there any other way to find out if current is flowing?
- Is it possible to make temporary magnets? How can these be made?
- How is heat generated in various electrical appliances?
- How do we know if a cell or a battery is dead? Can all cells and batteries be recharged?

This chapter establishes the deep link between **electricity and magnetism** (discovered by Oersted), explores **electromagnets**, then explains the **heating effect** of current, and finally looks inside **cells and batteries** — how they generate electricity.

4.1 Does an Electric Current Have a Magnetic Effect?

The chapter opens with a working model of a **lifting electromagnet**: an iron nail wrapped with wire, connected to a battery. When the circuit is closed, the nail picks up iron paper clips like a magnet; when opened, the clips fall off.

DEFINITION

When electric current flows through a conductor (like a wire), it produces a **magnetic field** around it. This is the **magnetic effect of electric current**. The magnetic field disappears when the current stops flowing.

4.1.1 Oersted's discovery

- **Hans Christian Oersted (1777–1851)**, a professor at a university in Denmark.
- In **1820**, while giving a demonstration, he noticed that whenever an electrical circuit was closed or opened, the needle of a magnetic compass lying nearby **deflected**.
- He investigated and, when certain that an electric current indeed produced a magnetic field, published his findings.
- This led to other scientists repeating his experiment and further investigating the connection between electricity and magnetism.

ACTIVITY 4.1 — KEY OBSERVATION

When current flows, the compass needle **deflects** from its original direction. When current stops, the needle **returns** to its original direction. The deflection indicates that the current-carrying wire has a magnetic effect on the compass needle.

DEFINITION

The region around a magnet or a current-carrying wire where its magnetic effect can be felt (e.g. by the deflection of a compass needle) is said to have a **magnetic field**.

4.1.2 Practical applications of the magnetic effect

- Electromagnets, electric bells, motors, fans, loudspeakers, and more.

4.1.3 Electromagnets

DEFINITION

A **current-carrying coil that behaves as a magnet** is called an **electromagnet**. For practical applications, most electromagnets have an **iron core** to make them stronger.

- When current flows through a coil, it behaves like a magnet and deflects a magnetic compass needle.
- When an **iron nail is inserted in the core**, the coil becomes a **stronger magnet** — deflection is much more, and it attracts iron clips.
- When current is stopped, the coil **loses its magnetic effect**.

4.1.4 Polarity of an electromagnet

- Just like a magnet, an electromagnet also has **two poles — North and South**.
- To find polarity: place a magnetic compass near one end (say A). The pole of the compass attracted to end A tells you the opposite pole is at A (if N of compass is attracted to A, then A is South).
- The polarity of end B is **opposite** to the polarity of end A.
- **Poles can be reversed** by changing the direction of the current.

4.1.5 Factors affecting the strength of an electromagnet

Factor	Effect on strength
Amount of electric current (more cells in battery)	More current → stronger magnetic field → more deflection, more clips attracted
Number of turns of the coil	More turns → stronger magnet
Iron core	Inserting an iron core makes the coil a much stronger magnet

- A single cell provides only a small current → weak magnetic field → less deflection, only a few clips.

4.1.6 Lifting electromagnets

- Strong electromagnets hung to cranes. The crane operator controls the magnet by switching current ON and OFF.
- Current ON → electromagnet lifts iron/steel objects. Current OFF → magnetic field disappears → objects are released.
- Widely used in **factories and scrap yards** to move, lift, and sort heavy metal items efficiently.

> **A STEP FURTHER — EARTH'S MAGNETISM**
> Earth itself behaves like a giant magnet. Deep inside the Earth, the movement of **liquid iron in the core** creates electric currents, which generate a magnetic field. Many migratory birds, fish, and animals use this field to navigate across continents and oceans. Earth's magnetic field also acts as a shield, blocking harmful particles from space.

4.2 Does a Current-Carrying Wire Get Hot?

> **DEFINITION**
> When an electric current passes through a conductor, it gets heated. This warming is known as the **heating effect of electric current**.

- When current flows through any conductor, it faces some opposition or **resistance** to its flow.
- Different conductors offer different levels of resistance.
- **Nichrome wire** offers higher resistance compared to a copper wire of the same size and length.
- This resistance causes some of the **electrical energy to be converted into heat energy**.

4.2.1 Factors on which heat generated depends

Factor	Effect
Magnitude of electric current	More current → more heat (2 cells heat the wire more than 1 cell for the same duration)
Material of the wire	Higher resistance (e.g. nichrome) → more heat

Thickness of the wire	Affects resistance and hence heat
Length of the wire	Affects resistance and hence heat
Duration for which current flows	Longer duration → more heat

> **SAFETY FIRST**
> Do not touch the nichrome wire for an extended period to avoid injuries. Hold only momentarily.

4.2.2 Applications of the heating effect

- **Incandescent lamp** — glows because its filament is heated by an electric current.
- Household appliances: **electric room heaters, electric stoves, electric kettles, electric irons, water heating immersion rods, hair dryers**.
- All these contain a rod or coil of wire called a **heating element**. In some appliances the element is visible and can be seen glowing red hot.

> **A STEP FURTHER**
> To prevent unnecessary heating in household switchboards, use appropriate wires, plugs, and sockets rated for the specified current. Overheating may damage plugs/sockets (plastic parts may melt) or even lead to fires. Household circuits have **safety devices** to minimise such incidents.

> **EVER HEARD OF... INDUSTRIAL APPLICATION**
> In **steel manufacturing industries**, a specially designed high-temperature **electric furnace** uses electric current to produce heat. This is used to melt and recycle scrap steel into usable steel.

4.3 How Does a Battery Generate Electricity?

4.3.1 Voltaic cell (Galvanic cell)

> **DEFINITION**
> A **Voltaic cell** (Galvanic cell) contains two metal rods made of different materials and a liquid called an **electrolyte**, placed in a glass or plastic container. A chemical reaction between the rods and the electrolyte produces electricity.

- Rods = **electrodes**; partly dipped in the electrolyte (usually a weak acid or salt solution).
- When the circuit is connected, current flows from the **positive terminal** through the circuit to the **negative terminal**.
- Over time, the chemicals get used up, and the cell stops working — it is then called **'dead'** and cannot supply any more electricity.

> **EVER HEARD OF... VOLTA & GALVANI**
> **Luigi Galvani** (late 1700s) noticed a dead frog's leg kicked when touched with two different metals — copper and iron. He thought the electricity came from the frog. **Alessandro Volta** believed it came from the metals. He used saltwater-soaked paper instead of the frog's leg and still got an electric current. This showed it was the **combination of metals and liquid** that generated current — leading to the invention of the first battery.

4.3.2 Lemon cell (Activity 4.6)

- Take 5–6 juicy lemons, copper wires/strips (1–2 mm thick), iron nails, one LED, connecting wires.
- Insert copper wire and iron nail in each lemon, keeping them apart.
- Join copper wires and nails as shown in Fig. 4.8b.
- Connect the LED between the copper wire of the first lemon and the iron nail of the last lemon.
- If the LED does not glow, **reverse its connections**.

- Metal electrodes = copper wires and iron nails; electrolyte = lemon juice.
- Salt solutions can be used instead of lemon juice.

> **REMEMBER**
> Current passes through an LED only when the **positive terminal (longer wire)** of the LED is connected to the **positive terminal** of the battery, and the **negative terminal (shorter wire)** to the **negative terminal**.

4.3.3 Common metal pairs for Voltaic cells

- Zinc/copper, zinc/silver, aluminium/copper, iron/copper, magnesium/copper, lead/copper.
- Some metals — like **copper** — act as **positive electrodes**; others — like **zinc** — act as **negative electrodes** (due to their chemical properties).

4.3.4 Dry cells

> **DEFINITION**
> **Dry cells** are widely used electric cells in which the electrolyte is **not a liquid but a thick moist paste**.

Part	Role
Zinc container	Acts as the **negative terminal**
Carbon rod at the centre	Covered with a metal cap that acts as the **positive terminal**
Paste-like electrolyte	Surrounds the carbon rod

- Dry cell is a **single-use cell** — once used up, it has to be disposed of.

4.3.5 Rechargeable batteries

- Can be **recharged and reused multiple times**; prevents wastage and saves money over time.
- Used in watches, phones, laptops, tablets, cameras, inverters, and electric vehicles.
- Rechargeable batteries also do **not last forever** — after many charge-use cycles, they slowly wear out. (This is why a phone battery requires charging more often after a year or two.)

> **A STEP FURTHER — E-WASTE**
> Even when a battery stops working, it is not completely 'dead'. It may still contain acids and metals like lead, cadmium, nickel, or lithium, which can cause fires or harm the environment if thrown in regular garbage. Many materials are valuable and can be recycled. Use special **'e-waste' recycling facilities**.

★ KEY TAKEAWAYS

- Electric current produces a **magnetic field** — discovered by **Oersted (1820)**.
- **Electromagnet** = current-carrying coil behaving as a magnet; iron core makes it stronger.
- Strength increases with **more current** and **more turns**; poles reverse when current direction reverses.
- **Heating effect**: current through a conductor produces heat due to **resistance**; nichrome has high resistance and is used in heating elements.
- Heat depends on current, material, thickness, length of wire, and duration.
- **Voltaic cell** = two different metal electrodes + electrolyte → chemical reaction produces electricity.
- **Dry cell** = zinc container (negative) + carbon rod (positive) + paste electrolyte; single-use.
- **Rechargeable batteries** can be reused but wear out over time.

CHAPTER 05

Exploring Forces

Grade 8 · Science

PROBE & PONDER

- Why does it feel harder to pedal a bicycle when going uphill than on flat ground?
- Why is it easier to slip on a wet surface?
- Why do we feel 'light' or like we are 'floating' just after our swing reaches its highest point and begins to come down?

This chapter defines **force**, explores what a force can do to an object, distinguishes **contact forces** (muscular, friction) from **non-contact forces** (magnetic, electrostatic, gravitational), and introduces **weight and its measurement**, plus **floating and sinking**.

5.1 What Is a Force?

DEFINITION

A **force** is a push or pull on an object resulting from the object's interaction with another object.

- The SI unit of force is **newton (N)**.
- At least **two objects must interact** for a force to come into play.

5.2 What Can a Force Do to Objects?

The force applied on an object may:

- Make an object **move from rest**.
- **Change the speed** of an object if it is moving.
- **Change the direction of motion** of an object.
- Bring about a **change in the shape** of an object.

Action	Push/Pull	Effect
Friend holding your moving bicycle from behind to stop it	Pull	Stopping or decreasing the speed of the bicycle
Hitting a moving ball with a bat	Push	Changing the direction of a moving ball
Pressing an inflated balloon	Push	Change in shape of the balloon

A STEP FURTHER

If an object is at rest, it does **not** mean no force is acting on it — it means the forces acting on the object are **balancing one another** (balanced forces).

5.3 Are Forces an Interaction Between Two or More Objects?

- When you push a table, your hand is one object applying force on another object — the table. They are **interacting**.
- Forces result only when **two objects interact** in some way.

5.4 Types of Forces

5.4.1 Contact forces

DEFINITION

> Forces that act only when there is **physical contact** between objects are called **contact forces**. Contact can be direct (hands, body parts) or indirect (a stick or rope).

Muscular force

- The force resulting due to the **action of muscles** in our body is known as muscular force.
- Occurs when muscles **contract and elongate** while doing any activity.
- Examples: walking, running, lifting, pushing, jumping, stretching.
- Animals, birds, fish, and insects use muscular forces for movement and survival.
- Humans have used the muscular force of animals (bullocks, horses) to carry out many tasks for a long time.

> **EVER HEARD OF...**
> Muscular force plays an important role **inside our body** too. It helps us chew food and push it through the alimentary canal during digestion. The expansion and contraction of our **heart muscles** allows blood to circulate — essential for survival.

Friction

> **DEFINITION**
> The force that comes into play when an object moves or tries to move over another surface is called the **force of friction** (or simply friction). Friction always acts in a direction **opposite to the direction of motion**. It is a contact force since it arises due to two surfaces in contact.

- A ball rolling on flat ground stops on its own; a bicycle slows down and stops if we stop pedalling.
- On a rough road, objects stop **sooner** than on a smoother road.
- Friction arises due to **irregularities** in the two surfaces in contact. Even surfaces that appear smooth have a large number of minute irregularities.
- When placed in contact, the irregularities of two surfaces **lock into each other** and oppose any effort to move one surface over the other.
- **Friction is greater on rough surfaces** (depends on the nature of the surfaces in contact).

> **A STEP FURTHER**
> Friction also acts when objects move through **liquids and gases**. Air, water, and other liquids exert force of friction on objects moving through them. Hence aeroplanes, ships, boats, and high-speed trains are designed with specific shapes to **reduce** the force of friction due to air or water around them.

5.4.2 Non-contact forces

> **DEFINITION**
> Forces whose effect can be experienced even if the objects are **not in contact** are called **non-contact forces**.

Force	Nature	Example
Magnetic force	Attractive or repulsive (like poles repel, unlike poles attract)	Compass needle pointing North; ring magnets floating on a stick
Electrostatic force	Attractive or repulsive (like charges repel, unlike charges attract)	Charged plastic scale attracting small paper pieces; rubbed balloons repelling each other
Gravitational force	Always **attractive** only	Objects falling towards the Earth

Magnetic force — the force exerted by a magnet on another magnet or a magnetic material.

- A magnet can exert force on another magnet without being in contact with it.

Electrostatic force

- When two objects of certain materials are **rubbed together**, electrical charges build up on their surfaces.

- These charges are called **static charges** as they do not move by themselves. The object that acquires static charges is said to be a **charged object**.
- A charged object **attracts** uncharged objects made of certain materials (e.g. small pieces of paper).
- The force exerted by a charged body on another charged body or an uncharged body is called **electrostatic force**. It is a non-contact force.
- **Like charges repel** each other; **unlike charges attract** each other.
- The two kinds of static charges are **positive** and **negative**.

> **A STEP FURTHER**
> When charges move, they constitute an **electric current** in an electrical circuit. It is the same current that makes a lamp glow or generates a heating effect or a magnetic effect.

Gravitational force

- All objects fall towards the Earth → the Earth attracts (pulls) them.
- The force with which the Earth attracts objects towards itself is called the **gravitational force**. Also called **force of gravity** or simply **gravity**.
- Since it acts without contact, it is a **non-contact force**.
- Gravitational force is always an **attractive force**.

> **VERTICAL MOTION**
> When an object is thrown vertically upwards: it moves up straight, slows down, stops momentarily at the top, then takes a straight vertical path downwards. While going up, speed decreases; while coming down, speed increases. This is **vertical motion** under the influence of gravitational force.

5.5 Weight and Its Measurement

> **DEFINITION**
> The force with which the Earth pulls an object towards itself is called the **weight** of the object. It measures how strongly an object is pulled by the Earth. SI unit of weight is **newton (N)**.

Mass	Weight
Amount of matter in an object	Gravitational force with which Earth (or another planet) pulls an object
Measured in **grams (g) or kilograms (kg)**	Measured in **newtons (N)**
Remains the same at every place	Can vary slightly from place to place on Earth; very different on different planets

> **A STEP FURTHER — WEIGHT ON DIFFERENT WORLDS**

Body	Mass	Weight
Earth	1 kg	10 N
Moon	1 kg	1.6 N
Mars	1 kg	3.8 N
Venus	1 kg	9 N
Jupiter	1 kg	25.4 N

> Mass remains the same; weight changes.

5.5.1 Spring balance

- A simple device used to measure **weight (force)**.

- Consists of a spring fixed at one end, with a hook at the other. When an object is hung from the hook, the spring stretches; the amount of stretching gives the weight.
- There is a scale marked to show weight in **newton**; usually also another scale showing corresponding **mass in gram (g)**.
- The mass scale assumes the spring balance is used on Earth.

Reading a spring balance:

- Maximum weight it can measure = its **range** (e.g. 0 to 10 N).
- Smallest value it can read = value of one small division = (difference between two bigger marks) ÷ (number of small divisions between them).
- Example: between 0 and 1 N, if there are 5 divisions, each small division = 1 N ÷ 5 = **0.2 N**.

> **A STEP FURTHER**
> In everyday language, the term 'weight' is often used when 'mass' is meant (e.g. "weight of the wheat bag is 10 kg"). In scientific use, this is not correct — mass is measured in g/kg, weight in newtons. Always use correct terms with correct units.

5.6 Floating and Sinking

> **DEFINITION**
> The force applied by a liquid on an object in the **upward direction** is known as **upthrust** or **buoyant force**.

- When an object is placed in a liquid, the gravitational force due to the Earth acts on it **downwards**.
- A buoyant force is applied on it by the liquid in the **upward direction**.
- If gravitational force > buoyant force → object **sinks**.
- If gravitational force = buoyant force → object **floats**.
- One factor on which buoyant force depends is the **density of the liquid**.
- An empty bottle pushed into water feels an upward push and bounces back to the surface.
- A mug feels lighter when it is inside water.

> **ARCHIMEDES' PRINCIPLE**
> When an object is fully or partially immersed in a liquid, it experiences an upward force which is equal to the **weight of the liquid it displaces**.
>
> - If weight of liquid displaced < weight of object → object **sinks**.
> - If weight of liquid displaced = weight of object → object **floats**.

> **EVER HEARD OF... PUMICE**
> Some rocks can float on water. One such rock is **Pumice**, formed during volcanic eruptions. When lava with lots of gas and water vapour cools quickly, it traps tiny bubbles of gas inside. This creates a light, porous rock filled with air pockets which is **less dense than water** and floats on it.

★ KEY TAKEAWAYS

- **Force** = push or pull resulting from interaction between two objects; SI unit = **newton (N)**.
- Force can: move a resting object, change speed, change direction, change shape.
- **Contact forces** — muscular force, friction. **Non-contact forces** — magnetic, electrostatic, gravitational.
- **Friction** opposes motion; arises from surface irregularities; greater on rough surfaces.
- **Gravitational force** is always attractive. **Magnetic** and **electrostatic** forces can be attractive or repulsive.
- **Weight** = force with which Earth pulls an object (newtons); **mass** = amount of matter (g/kg) — mass is constant, weight varies by location.
- **Spring balance** measures weight; smallest division gives its accuracy.
- **Buoyant force (upthrust)** acts upward on objects in a liquid; Archimedes' Principle explains floating and sinking.

CHAPTER 06

Pressure, Winds, Storms, and Cyclones

Grade 8 · Science

PROBE & PONDER

- Why are winds stronger on some days than on others?
- Why are water tanks usually placed at a height?
- Can air pressure really crush us?
- What causes storms and cyclones?
- If the Earth stopped rotating, would cyclones still form?

This chapter builds from the basic idea of **pressure** (force per unit area) to **liquid pressure**, **atmospheric pressure**, the **formation of wind**, and finally the dramatic weather events — **thunderstorms, lightning, and cyclones**.

6.1 Pressure

DEFINITION & FORMULA

Pressure = Force ÷ Area
SI unit of force = newton (N); SI unit of area = metre² (m²).
Therefore, SI unit of pressure = **newton/metre² (N/m²)**, also called a **pascal (Pa)**.

- Consider only forces that act **perpendicular** to the surface.
- Broad straps reduce the pressure exerted by a bag on our shoulders (larger area) compared to narrow straps.
- Easier to lift a water-filled bucket with a broad handle than a narrow handle.
- People carrying loads on their heads place a round piece of cloth under the load — to reduce pressure by increasing area.
- Example calculation: Force = 100 N on area 2 m² → Pressure = 100/2 = **50 N/m²**.

Activity	Mode of action	Easy/Difficult?	Reason
Driving an iron nail	By the head of the nail	Difficult	Larger area → less pressure
Driving an iron nail	By the pointed end	Easy	Smaller area → more pressure
Cutting an apple	Sharp edge of knife	Easy	Smaller area → more pressure
Cutting an apple	Blunt edge of knife	Difficult	Larger area → less pressure

6.1.1 Pressure exerted by liquids

- Liquids exert pressure not only at the bottom of the container but also **on its sides**. In fact, liquids exert pressure **in all directions**.
- Pressure exerted by a liquid in a vessel depends on the **height of its column**.
- As height increases, pressure at the bottom increases.
- This is why **overhead tanks are placed at a height** — to increase pressure in the taps so water flows in a good stream.
- A person on a lower floor gets a more powerful stream than one on a higher floor (greater height difference = greater pressure).

EVER HEARD OF... DAMS

The base of a dam is much broader than the top. A broad base not only supports the structure but also withstands the **horizontal water pressure** near the bottom. The pressure which acts horizontally is very large near the bottom — hence the base is made broader.

6.2 Pressure Exerted by Air

DEFINITION

The envelope of air surrounding the Earth is called the **atmosphere**. It contains nitrogen, oxygen, argon, carbon dioxide, and other gases in small quantities. It extends up to many kilometres above the Earth's surface.

DEFINITION

The pressure exerted by air around us is known as **atmospheric pressure**.

- Air exerts pressure on **all objects**.
- Air exerts pressure **in all directions** — that is why a balloon expands in all directions when filled.
- When an inflated balloon is left open, air escapes because the pressure inside is higher than outside.
- Force exerted by the atmospheric air column over an area 15 cm × 15 cm is nearly equal to the force of gravity on an object of mass **225 kg (2250 N)**.
- We are not crushed because the pressure **inside our bodies** is also equal to atmospheric pressure — this balances the pressure from outside. The internal pressure is caused by movement of fluids and gases in tissues and organs.

6.2.1 Rubber sucker

- When pressed against a smooth flat surface, most of the air between the cup and the surface is pushed out → air pressure inside is reduced.
- The sucker sticks because the pressure of air surrounding the sucker is **higher** than the pressure inside.
- To pull it off, the applied force must overcome the **pressure difference**.

A STEP FURTHER — UNITS

Practical unit of air pressure = **millibar (mb)** = 100 Pa. Also expressed in **hectopascal (hPa)** = 100 Pa.

6.3 Formation of Wind

KEY PRINCIPLE

Air moves from a region of high air pressure to a region of low air pressure.

- Air flow continues until the pressure in both regions becomes **equal**.
- The **speed** of air is higher if the **pressure difference is higher**.

6.3.1 Sea breeze and land breeze

Breeze	Time	What happens
Sea breeze	Day	Land heats faster than water → air above land rises → low pressure over land → air from high-pressure sea blows to land
Land breeze	Night	Water is warmer than land → low pressure over sea → wind blows from land to sea

6.4 High-Speed Winds Result in Lowering of Air Pressure

KEY PRINCIPLE

High-speed winds are accompanied by a reduced air pressure.

- When you blow between two hanging balloons, they move towards each other — a low-pressure area is created between them, and higher surrounding pressure pushes them together.
- Blowing harder → balloons approach faster.
- When high-speed winds blow over houses, a low-pressure area is created over the roof. If the pressure difference is large and roofs are weak, they may be blown away.
- It is **safer to keep doors and windows open** during storms with high-speed winds — the wind then moves through the house too, reducing the pressure difference and preventing roofs from being blown off.

6.5 Storms, Thunderstorms, and Lightning

6.5.1 How a storm forms

- Land gets heated → warm and moist air, being lighter, **rises**, creating a low-pressure area.
- Cooler air from surrounding high-pressure areas flows in to take its place. This air, in turn, gets heated and rises — a continuous process of **wind circulation**.
- As the rising air expands, it **cools** and moisture in it **condenses** to form water droplets → **clouds**.
- Water droplets merge to form heavier drops, which come down as **rain, hail, or snow**.
- **Storm** = strong winds accompanied by rain.
- In hot, humid, and tropical regions like India, storms are more frequent.

6.5.2 How lightning forms

- Under certain conditions, warm air rises to great heights where low temperature converts water droplets into **ice particles**.
- Strong winds blowing **upwards and downwards** cause rubbing between water droplets and ice particles.
- Rubbing causes **static electric charges** to develop within the clouds.
- **Positively charged** lighter ice particles move upwards (upper part of clouds).
- **Negatively charged** heavier water droplets occupy the lower part of the clouds.
- When the negatively charged lower part of the cloud moves closer to the ground, it causes the ground and nearby objects (trees, buildings) to become **positively charged**.
- Normally, air acts as an **electrical insulator** and does not let opposite charges meet. When the built-up charge becomes very large, the insulating property of air **breaks down**.
- A sudden flow of charges takes place, producing a bright flash of light called **lightning**.
- Lightning can occur: within a cloud, between clouds, or between clouds and the ground.
- Lightning **rapidly heats** the air around it, causing the air to expand and produce a loud sound called **thunder**.
- A storm accompanied by lightning and thunder is called a **thunderstorm**.

> **A STEP FURTHER — LOCAL THUNDERSTORMS OF INDIA**
>
> - **Kalboishakhi** — West Bengal, Bihar, Jharkhand. Occur before the arrival of the monsoon; help kharif crops grow.
> - **Bordoila** — Assam. Occur before the monsoon.
> - **Mango showers** — Kerala, Karnataka, Tamil Nadu. Support the ripening of mangoes.
> - Local thunderstorms in Karnataka help in the growth of **coffee plants**.

6.5.3 Safety during lightning

- Stay away from tall objects.
- Find a low-lying open area and **crouch down**.
- Minimise contact with the ground.
- Do **not** lie down flat.
- Avoid using an umbrella with a metallic rod.

- If you are in water, get out of it.
- If inside a bus or a car, you are comparatively safer.

6.5.4 Lightning conductor

- A **metallic rod** installed along the walls of buildings during construction.
- One end is **pointed** and kept higher than the highest point of the building.
- The other end is buried **deep in the ground**.
- Provides an **easy path** for the transfer of electric charges into the ground.

6.6 Cyclone

> **DEFINITION**
> A **cyclone** is a large storm that forms over warm ocean waters — a spinning system of clouds, winds, and rain with a very low-pressure area at the centre.

How a cyclone forms:

- Ocean water gets heated → warm and moist air above it rises.
- As moist air rises, water vapour **condenses** to form raindrops.
- During evaporation, water takes up heat; when vapour condenses, heat is **released back** into the atmosphere.
- This causes further warming of the ascending air → it rises even further → creates an even **lower pressure**.
- Air from surrounding regions rushes in and also starts rising.
- **Earth's rotation** causes the moving air to **spin**.
- This cycle repeats, resulting in a very low-pressure area with high-speed winds revolving around it.

6.6.1 Structure of a cyclone

- **Eye of the cyclone** — the region of lowest pressure at the centre. At the eye, the wind is **calm**.
- Surrounding region — strong winds and heavy rainfall.
- As a cyclone moves from the ocean towards land, it generates **higher wind speeds** than regular thunderstorms.
- Once it reaches land, the source of moist air is cut off, and it gradually **loses strength**.

6.6.2 Destructive effects of cyclones

- Leaves behind a trail of destruction that can take months or years to repair.
- Example: **Amphan cyclone (2020)** had peak wind speeds of **270 km/h**.
- Strong winds push ocean water towards the shore, creating a **wall of water 3–12 metres high** (storm surge).
- Floods coastal areas and even areas far from the sea.
- Heavy rainfall may cause rivers to overflow and trigger **landslides**.
- Seawater rushing inland **contaminates drinking water** and damages farmland; salt makes soil less fertile.
- Roads blocked by fallen trees and debris; power outages can last for days.

6.6.3 Protection during cyclones

- Stay updated on weather reports and periodic alerts/warnings issued by the **India Meteorological Department (IMD)**.
- Weather monitoring satellites help track cyclones and predict their path.
- If in a cyclone-prone area, keep an **emergency kit** ready.
- During a cyclone, move quickly to a nearby designated **cyclone shelter**.

★ KEY TAKEAWAYS

- **Pressure = Force / Area**; SI unit = **pascal (Pa)** = N/m².
- Smaller area → higher pressure (nail point, knife edge).
- Liquid pressure depends on **height of the column**; acts in all directions.
- **Atmospheric pressure** acts on all objects in all directions; balanced by internal body pressure.
- **Air moves from high pressure to low pressure**; greater pressure difference → higher wind speed.
- **High-speed winds → lower air pressure** (why roofs blow off; keep doors/windows open during storms).
- Lightning = sudden flow of charges between cloud and ground/cloud; thunder = sound from rapid heating of air.
- **Cyclone** = spinning system over warm ocean; **eye** is calm, surrounded by strong winds and rain.

CHAPTER 07

Particulate Nature of Matter

Grade 8 · Science

PROBE & PONDER

- Why is it possible to pile up stones or sand, but not a liquid like water?
- Why does water take the shape of folded hands but lose that shape when released?
- We cannot see air, so how does it add weight to an inflated balloon?
- Is the air we breathe today the same that existed thousands of years ago?

This chapter builds the core idea that **matter is made of tiny particles**, explains the three **states of matter** in terms of interparticle spacing and interparticle attraction, and describes how particles **move** in each state.

7.1 What Is Matter Composed Of?

- Rocks in mountains break down by **erosion**; rivers carry the pieces, breaking them further into pebbles, stones, sand, and clay.
- Even the finest grain of sand or chalk powder is not the smallest unit — it is made of a large number of **constituent particles**.

DEFINITION

A **constituent particle** is the basic unit that makes up a larger piece of a substance or material.

- When sugar dissolves in water, it breaks up into its constituent particles, which **cannot be seen** but can be sensed by taste.
- The tiny sugar particles separate and occupy the available spaces between the water particles. These spaces are called **interparticle spaces**.
- Matter is composed of a large number of extremely small particles — so small they cannot be seen even through an ordinary microscope.

OUR SCIENTIFIC HERITAGE — ACHARYA KANAD

An ancient Indian philosopher who first spoke about the idea of a **Parmanu (atom)**. He believed matter is made up of tiny, indivisible eternal particles called Parmanu. This idea was written in his work called the **Vaisheshika Sutras**.

7.2 What Decides Different States of Matter?

DEFINITION

The constituent particles of matter are held together through forces which are **attractive** in nature. These forces are called **interparticle attractions**. The strength of these attractions depends on the nature of the substance and the interparticle distance. Even a slight increase in distance decreases the interparticle forces drastically. The strength of these forces ultimately decides the **physical state** of the substance.

7.2.1 Solid state

- Solids have a **definite shape and volume**.
- Particles are **tightly packed**; interparticle attractions are **very strong**.
- Strong forces hold particles in **fixed positions**, preventing them from moving freely.
- Particles can only move to and fro about their positions — they **vibrate or oscillate** but cannot move past each other.
- When solids are heated, particles vibrate more vigorously. At a stage, they start leaving their positions; interparticle forces weaken; the solid converts into a liquid. This temperature is the **melting point**.

DEFINITION

The minimum temperature at which a solid melts to become a liquid at atmospheric pressure is called its **melting point**.

Material	Melting point
Ice	0 °C
Urea	133 °C
Iron	1538 °C

Note: Some solids have weak interparticle forces (low melting points); others have strong forces (high melting points). **Ice is an exception** — its particles are farther apart than in water.

7.2.2 Liquid state

- Liquids **do not have a fixed shape** and take the shape of the container.
- Liquids have a **definite volume** — the level remains the same when poured into different containers (200 mL stays 200 mL).
- Particles of liquids are **free to move** but only within a limited space.
- Interparticle attractions are **slightly weaker than in solids** but still strong enough to keep particles close together.
- You can move your finger through water without breaking or cutting it permanently — water is temporarily displaced and returns to position.
- When a liquid is heated, a stage comes when it starts boiling. The temperature at which a liquid boils and turns into vapour at atmospheric pressure is its **boiling point**.
- At the boiling point, vapour formation is fast and occurs **within** the liquid (bubbles).
- **Evaporation** — slower vapour formation at all temperatures, occurring only at the surface.

7.2.3 Gaseous state

- Gases **do not have a fixed shape or volume** — they occupy the entire available space.
- Particles move freely in all directions; interparticle attractions are **negligible**.
- Like liquids, gases acquire the shape of the vessel they are in.
- Both liquids and gases **flow** and do not retain a fixed shape — these properties classify them as **fluids**.

SMOKE / IODINE VAPOUR DEMONSTRATION

Smoke from an incense stick fills the entire gas jar, showing that gas particles move freely and spread to fill all available space. Iodine vapour can also be used for the same demonstration.

7.3 How Does Interparticle Spacing Differ in the Three States?

Property	Solid	Liquid	Gas
Interparticle spacing	Minimum	Little more than in solids	Maximum
Packing	Closely packed	Loosely packed than solids	Particles are free
Interparticle attraction	Maximum	Slightly weaker than in solids	Minimum (negligible)
Movement of particles	Negligible (only vibrations)	Restricted to limited space	Free, in all available space
Shape	Definite	Takes container's shape	Takes container's shape
Volume	Definite	Definite	Not fixed
Compressibility	Practically incompressible	Practically incompressible	Highly compressible

ACTIVITY 7.6 — SYRINGE TEST

Pushing the plunger of a syringe (with the open end closed by thumb) compresses the air inside — volume decreases. This shows gas particles have a lot of space between them, which can be reduced by applying external pressure. **Water is practically incompressible** in the same test.

ACTIVITY 7.7 — SUGAR IN WATER

When sugar is added to water, the level initially rises; after dissolution it may decrease. Since the volume of the solution is **less than the sum of the volumes of water and sugar**, it indicates there is space between water particles — the dissolved particles occupy these spaces.

- **Sugar dissolves** because water particles can pull sugar particles apart.
- **Sand does not dissolve** — its constituent particles are held together too strongly for water particles to pull apart.

7.4 How Particles Move in Different States of Matter

- Water particles are in **constant motion**.
- When a grain of potassium permanganate is added to water, streaks of pink colour spread out; eventually the entire bulk becomes uniformly pink.
- Water particles first pull out the particles of potassium permanganate, then hit them so they spread throughout the liquid.
- Movement of particles **increases when heat is provided**.
- Potassium permanganate spreads fastest in **hot water**, less quickly at room temperature, slowest in **ice-cold water**.
- When an incense stick is burnt, the fragrance spreads throughout the room — air particles hit fragrance particles and help them spread.

EVER HEARD OF... SOAP

The particulate nature of matter plays a role in everyday processes. When washing clothes stained with oil, numerous soap particles surround the oil particles. One end of the soap particle attaches to the oil, and the other mixes with water — helping lift the oil off and wash it away.

SUMMARY — WHY THE STATE DEPENDS ON THERMAL ENERGY

Matter is made of small particles held together by attractive forces. The strength of these forces depends on the distance between particles, which in turn depends on their **thermal (heat) energy**.

- **Solid**: low thermal energy → particles close → strong attractions → only vibrations.
- **Melting point**: thermal energy overcomes attractive forces → particles can move away from fixed positions → liquid.
- **Liquid**: particles move around within limited space.
- **Gas**: particles have enough energy to overcome attractions → move freely in all directions.

★ KEY TAKEAWAYS

- Matter is made of tiny **constituent particles** with **interparticle spaces** between them.
- The **strength of interparticle attraction** and the **interparticle distance** decide the physical state.
- Solid = definite shape and volume; liquid = definite volume but no fixed shape; gas = neither fixed shape nor fixed volume.
- Solid particles vibrate only; liquid particles move within limited space; gas particles move freely in all directions.
- **Melting point** = temperature at which a solid becomes liquid; **boiling point** = temperature at which a liquid becomes vapour.
- Gases are highly compressible; solids and liquids are practically incompressible.
- Movement of particles increases with heat — the basis of all state changes.
- **Acharya Kanad** proposed the idea of the **Parmanu (atom)** in the Vaisheshika Sutras.

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GRADE 8 · SCIENCE

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- Tables compare and classify — memorise the rows, they are common exam questions.
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CHAPTER 08

Nature of Matter: Elements, Compounds, and Mixtures

Grade 8 · Science

PROBE & PONDER

- Which entities consist of matter, and which do not?
- How can elements be combined to form a compound?
- How could a compound that absorbs CO_2 from air contribute to solving environmental challenges?

This chapter classifies matter into **mixtures** (uniform and non-uniform) and **pure substances** (elements and compounds), then extends the idea to **minerals** and everyday materials.

8.1 What Are Mixtures?

DEFINITION

When two or more substances are mixed, where each substance **retains its properties**, it is called a **mixture**. The individual substances that make up a mixture are called its **components**. The components do **not react chemically** with each other.

Type	Description	Examples
Non-uniform mixture	Components are generally visible with the naked eye or a magnifying device	Sprout salad, poha, sand and water, sawdust and water
Uniform mixture	Components are evenly distributed and cannot be distinguished, even with a microscope	Sugar and water, salt and water, air

A STEP FURTHER — ALLOYS

Stainless steel is a mixture of iron, nickel, chromium, and a small amount of carbon — mixed so uniformly that it appears the same throughout. Such mixtures are called **alloys**. **Brass** = copper + zinc. **Bronze** = copper + tin.

OUR SCIENTIFIC HERITAGE — MISHRA LOHA

Ancient Indian texts (Charaka Samhita, Susruta Samhita, Rasaratna Samucchaya, Rasa Jala Nidhi) mention the use of alloys for medicinal purposes. **Bronze (Kamsya)** = Copper (Tamra, 4 parts) + Tin (Vanga, 1 part); used to improve digestion and boost immunity.

8.1.1 Is air a mixture?

- Air is a **uniform mixture** of mainly **nitrogen (~78%)**, oxygen, argon, carbon dioxide, and water vapour.
- Oxygen is required by most living beings to stay alive and helps in combustion.
- Nitrogen does not take part in combustion.
- Water vapour in air condenses into liquid water on a cool surface.
- **Lime water test**: Calcium oxide (quick lime) + water → calcium hydroxide (lime water). Lime water turns milky when carbon dioxide reacts with it to form calcium carbonate (insoluble white particles) and water — demonstrating the presence of CO_2 in air.
- Dust particles are suspended in air — they are **pollutants**, not an integral part of air.
- **Air Quality Index (AQI)** — a tool used to describe air quality.

8.1.2 Types of mixtures (Table 8.1)

Mixture type	Examples	Uniform or non-uniform
Gas + gas	Air	Uniform

Gas + liquid	Aerated water (soda water); oxygen dissolved in water	Uniform
Solid + gas	Carbon particles in air	Non-uniform
Liquid + liquid	Acetic acid in water (vinegar); oil and water	Vinegar: uniform; Oil + water: non-uniform
Solid + liquid	Sand and water; seawater	Sand + water: non-uniform; Seawater: uniform
Solid + solid	Baking powder (baking soda + tartaric acid); alloys	Uniform

8.2 What Are Pure Substances?

- In **common usage**, 'pure' means unadulterated products. **Adulteration** is the illegal process of adding cheaper or poor-quality substances to a product.
- In **science**, a pure substance is one that has **no other substance present in it**.

> **DEFINITION**
> A **pure substance** is a kind of matter that cannot be separated into other kinds of matter by any physical process. It consists of the **same type of particles**.

8.3 Types of Pure Substances

8.3.1 Elements

> **DEFINITION**
> **Elements** are substances that cannot be further broken down into simpler substances. They are the building blocks of all matter. Each element is made up of identical particles called **atoms**.

- Examples: gold, silver, sulfur, carbon, hydrogen, oxygen.
- Atoms of most elements cannot exist independently. Two or more atoms combine to form a stable particle called a **molecule**.
- Examples: two atoms of hydrogen → one molecule of hydrogen; two atoms of oxygen → one molecule of oxygen.
- Elements are classified into **metals** and **non-metals**.
- **Metalloids** (e.g. silicon, boron) have intermediate properties between metals and non-metals.

> **A STEP FURTHER — ELEMENT FACTS**
> - Number of elements known at present = **118**.
> - Most exist in a solid state.
> - **11 elements** exist in a gaseous state at room temperature — all non-metals (oxygen, helium, nitrogen, etc.).
> - Only **2 elements are liquid** at room temperature — **mercury** (a metal) and **bromine** (a non-metal).
> - Gallium and caesium are solid but become liquid around **30°C (303 K)**.

8.3.2 Compounds

> **DEFINITION**
> **Compounds** are formed when different elements combine in **fixed ratios** to form something entirely new. The properties of compounds are **different** from those of the elements forming them. The constituent elements of a compound **cannot be separated by any physical method**.

- **Water** — molecules made of two different elements: hydrogen and oxygen, combined chemically in a fixed ratio of **2:1**.
- **Sodium chloride (common salt)** — sodium (soft metal) + chlorine (hazardous gas) combine in a **1:1** ratio to form a harmless, taste-enhancing substance essential for life.

- **Sugar** — on heating, decomposes to give **carbon** and **water**. Water consists of hydrogen and oxygen. Hence sugar is a compound of **carbon, hydrogen, and oxygen**.

8.3.3 Comparison — mixture vs. compound

Feature	Mixture	Compound
Combination	Physical mixing	Chemical combination
Ratio	Not fixed	Fixed ratio
Properties	Components retain their properties	Properties are different from constituent elements
Separation	Can be separated by physical methods	Cannot be separated by physical methods
Examples	Air, seawater, brass	Water, sodium chloride, sugar, iron sulfide

8.4 How Do We Use Elements, Compounds, and Mixtures?

- Air we breathe = mixture of gases (oxygen, nitrogen, carbon dioxide).
- Water = compound of hydrogen and oxygen.
- Elements like iron and aluminium = used to construct bridges, buildings, vehicles.
- Chemists study how elements combine to create compounds → life-saving medicines and vaccines.
- Knowledge of compounds and mixtures helps create fertilisers for crop production.
- Engineers and material scientists design materials with unique properties — alloys like **stainless steel** (stronger and more durable than pure iron).
- Wood, steel, and concrete (building materials) are all **mixtures**.

> **A STEP FURTHER — GRAPHENE AEROGEL**
> An example of a 'wonder material' developed by material scientists. Made from **carbon**, it is said to be the **lightest material on Earth** — so light that even grass can hold it. Highly porous, high absorbing capacity. Potential uses: cleaning oil spills on seas and land, energy-saving devices, special coatings for buildings.

8.5 What Are Minerals?

- Most rocks are a mixture of **minerals**, viewable with the eyes or with a magnifying glass/microscope.
- **Native minerals** — pure elements, not compounds. Can be metals (gold, silver, copper) or non-metals (sulfur, carbon).
- Most minerals are **compounds** made up of more than one element.
- Examples of minerals: **quartz, calcite, mica, pyroxene, olivine, talc**.
- Everyday uses: cement made from calcite, quartz, alumina, iron oxide; talcum powder from talc.

> **OUR SCIENTIFIC HERITAGE — DHOKRA ART**
> An old craft from **Bihar and Odisha** that uses different metals to create beautiful figures inspired by nature. Process: shape design in beeswax → cover with clay to make a mould → wax melted out → hollow space filled with molten **brass or bronze** → strong, shiny golden colour. Figures show animals, people, nature, reflecting tribal creativity.

> **MATTER vs. NON-MATTER**
> Elements and compounds are the building blocks of matter — everything that has **mass** and takes up **space**. However, not everything is matter: **light, heat, electricity, thoughts, and emotions** are important parts of our world but are **not made of matter**.

★ KEY TAKEAWAYS

- **Mixture** = two or more substances mixed; components retain properties; no chemical reaction.
- **Uniform** mixtures (sugar + water, air) vs. **non-uniform** (sand + water, salad).
- **Alloys** (stainless steel, brass, bronze) are uniform mixtures of metals.
- **Pure substance** = same type of particles; cannot be separated by physical means.
- **Elements** = simplest substances, cannot be broken down further; made of atoms.
- **Compounds** = elements combined chemically in fixed ratios; properties different from constituents.
- Water = H:O = 2:1; NaCl = Na:Cl = 1:1; Sugar = C, H, O.
- **Minerals** = natural solids with fixed chemical composition; most are compounds, some are native elements.

CHAPTER 09

The Amazing World of Solutes, Solvents, and Solutions

Grade 8 · Science

PROBE & PONDER

- What happens when you add too much sugar to tea and it stops dissolving? How can you solve this?
- Why do sugar and salt dissolve in water but not in oil? Why is water considered a good solvent?
- Why are water bottles usually tall and cylindrical instead of spherical?

This chapter explores **solutions** — uniform mixtures of solute and solvent — and then moves to **solubility**, the effect of temperature, **density**, and why objects **float or sink**.

9.1 What Are Solute, Solvent, and Solution?

DEFINITIONS

A **solution** is a uniform mixture (e.g. salt or sugar and water).
Solute — the solid component that dissolves (or, in liquid-liquid solutions, the substance present in **smaller amount**).
Solvent — the liquid component that dissolves the solute (or, in liquid-liquid solutions, the substance present in **larger amount**).

Solute + Solvent → Solution

- When a solution is formed by mixing two liquids, the substance present in smaller amount is the solute; the one in larger amount is the solvent.
- Air is a mixture of gases — it can also be considered a solution (uniform mixture of gases).

9.2 How Much Solute Can a Fixed Amount of Solvent Dissolve?

DEFINITIONS

Unsaturated solution — a solution in which more solute can be dissolved at a given temperature.
Saturated solution — a solution in which the solute stops dissolving and begins to settle at the bottom (at that particular temperature).
Concentration — the amount of solute present in a fixed quantity of solution (or solvent).
Dilute solution — less amount of solute. **Concentrated solution** — more amount of solute. (These are relative terms.)
Solubility — the maximum amount of solute that dissolves in a fixed quantity of the solvent.

- Initially, salt completely dissolves; after adding more, a stage comes when undissolved salt settles at the bottom — the solution is now **saturated** at that temperature.

9.2.1 Effect of temperature on solubility

- For **most substances**, solubility **increases** with an increase in temperature.
- A saturated solution at a particular temperature behaves as an **unsaturated solution** if the temperature is increased.
- Experiment: Baking soda dissolves more in water at 70°C than at 50°C; even less at 20°C.

OUR SCIENTIFIC HERITAGE — SOLVENTS IN AYURVEDA

Water has primarily been used as a solvent for medicinal formulations in Ayurveda, Siddha, and other traditional systems. Drug formulations have also been prepared using **hydro-alcoholic extracts** of herbs. Oils, ghee, milk, and other substances have been used as solvents to help achieve therapeutic benefits.

BE A SCIENTIST — ASIMA CHATTERJEE

Renowned for developing **anti-epileptic and anti-malarial drugs**. Used solvents and solutions extensively to extract and isolate important compounds from medicinal plants. Earned a Doctorate of Science — the second Indian woman to do so after Janaki Ammal. First woman to receive the **Shanti Swarup Bhatnagar Award** in chemical science; honoured with the **Padma Bhushan**.

9.3 Solubility of Gases

- Many gases, including **oxygen**, dissolve in water — only to a small extent.
- This dissolved oxygen sustains **all aquatic life** — plants, fishes, and other organisms.
- The mixture of gases in water is a **uniform mixture** (gases dissolve evenly).
- Solubility of gases **decreases** as temperature **increases**.
- More oxygen can dissolve in **cold water** — ensuring sufficient oxygen for aquatic life.
- When water warms up, solubility of oxygen decreases.

9.4 Why Do Objects Float or Sink in Water?

- Some objects float (husk, oil, wood) while others sink (rice, iron rod).
- It is generally believed that floating objects are lighter and sinking objects are heavier than the liquid.
- A wooden stick and an iron rod may be of the same size, yet the iron rod feels much heavier.
- When we say iron is heavier than wood, we refer to a property called **density**.

NOTE

The density of a substance is **not the only factor** that decides whether it will float or sink in a particular liquid.

9.5 What Is Density?

DEFINITION & FORMULA

Density is defined as the mass present in a unit volume of that substance.
Density = Mass / Volume

- Density of a substance is **independent of its shape or size**.
- However, it is **dependent on temperature and pressure**.
- Pressure primarily affects the density of **gases**; its effect on solids and liquids is negligible.
- SI unit of density = **kilogram per cubic metre (kg/m³)**.
- Other units: **g/mL** (gram per millilitre), **g/cm³** (gram per cubic centimetre).

Conversion factor for density

1 kg/m³ = 1000 g/m³ = 1000 g/1000 L = 1 g/L = 1 g/1000 mL = 1 g/1000 cm³

- The mass of **1 mL of water** is close to **1 g** at room temperature.
- Hence, 10 mL water ≈ 10 g; 100 mL water ≈ 100 g.
- Example: Aluminium block of mass 27 g and volume 10 cm³ → density = **2.7 g/cm³**.
- Aluminium is **2.7 times denser than water**. This is expressed as the **relative density** of aluminium with respect to water = 2.7 (a number without units).

RELATIVE DENSITY

Relative density of any substance with respect to water = Density of that substance / Density of water at that temperature

> **THINK LIKE A SCIENTIST**
> Some packets of ghee or oil are labelled with a volume of **1 litre** but a weight of only, say, **910 grams**. This tells us the density of the oil is **less than that of water** (since 1 L of water weighs about 1000 g).

9.5.1 Determination of density

How to measure mass:

- Mass is the quantity of matter present in any object. Measured using a **balance**.
- Digital weighing balance: switch ON → ensure zero reading (press tare/reset if needed) → place watch glass/butter paper → reset to zero → place solid object → note reading.
- For liquids: replace watch glass with a beaker and pour the desired amount.

How to measure volume:

- Volume = space occupied by an object. SI unit = **cubic metre (m^3)**.
- Smaller volumes expressed in **decimetre cube (dm^3)** or **centimetre cube (cm^3)**. 1 cm^3 = 1 cc.
- Volume of liquids expressed in **litres (L)** = 1 dm^3. Submultiple: **millilitre (mL)** = 1 cm^3.
- **Measuring cylinder** — narrow transparent cylindrical container with markings. Available in 5 mL, 10 mL, 25 mL, 50 mL, 100 mL, 250 mL, etc.

> **READING A MEASURING CYLINDER**
> - Maximum volume it can measure = its capacity (e.g. 100 mL).
> - Smallest volume it can measure = (difference between two bigger marks) ÷ (number of small divisions).
> - For a 100 mL cylinder: between 10 mL and 20 mL, there are 10 divisions → 1 small division = 1 mL.
> - **Meniscus** — the curved surface of water in the cylinder. For colourless liquids, read the mark that coincides with the **bottom** of the meniscus. For coloured liquids, read the **top** of the meniscus.
> - Keep eyes at level with the bottom of the meniscus while reading.

Determining volume of regular solids:

- For cuboid: **Volume = l × w × h**.
- Example: notebook 25 cm × 18 cm × 2 cm = **900 cm^3**.

Determining volume of irregular solids (Activity 9.7):

- Fill measuring cylinder with water to a known volume (e.g. 50 mL).
- Tie the object (stone) with a thread and lower it into the cylinder.
- Record the final volume (e.g. 55 mL).
- **Volume of object = Final volume − Initial volume** = 55 − 50 = **5 mL = 5 cm^3**.

> **LET US CALCULATE DENSITY**
> Density = Mass / Volume = 16.400 g / 5 cm^3 = **3.28 g/cm^3**

> **LET US DIG DEEPER — EARTH'S LAYERS**
> Earth is composed of several layers (crust, upper mantle, lower mantle, outer core, inner core), each with a particular range of density. The outermost layer, the **crust**, is the **lightest**; density **increases** as we move towards the centre. As one moves deeper, both **pressure** and **temperature** rise significantly, making materials heavier and more compact.

> **EVER HEARD OF... BAMBOO RAFTS**
> In ancient times, people used **bamboo** and **wooden logs** to travel across rivers and seas. Bamboo was used because it is **light, hollow, and floats easily**. People tied bamboo poles together to make rafts and small boats for fishing, trading, and crossing water bodies. Wooden logs were hollowed out to make boats or used as rafts.

9.5.2 Effect of temperature on density

- Generally, density **decreases** with heating and **increases** with cooling.
- As temperature increases, particles tend to move away and spread → volume increases → mass unchanged → **density decreases**.
- Hot air moves up because it is **less dense** than the cool air around it. Hot air balloons work on this principle.

9.5.3 Effect of pressure on density

State	Effect of increasing pressure
Gas	Particles move closer → volume decreases → **density increases**
Liquid	Small effect (nearly incompressible)
Solid	Even less affected than liquids; changes usually **negligible**

EVER HEARD OF... WHY ICE FLOATS ON WATER

Ice floats because it is **lighter than liquid water**. Water has a special property: its density is highest at **4°C**. As temperature drops and water turns into ice at **0°C**, it undergoes a change in structure — particles arrange themselves in a way that takes up **more space** (expansion). Same mass occupies larger volume → density decreases → ice floats. This is important for animals in lakes and oceans: ice forms a layer on top, keeping the water underneath warm enough for fish and other creatures to survive in extremely cold weather.

THINK LIKE A SCIENTIST — EGG IN WATER

A raw whole egg **sinks** in tap water. To make it float, you can add salt to the water (increasing the density of the liquid).

★ KEY TAKEAWAYS

- **Solution** = uniform mixture; **solute** + **solvent** → solution.
- **Unsaturated** = more solute can dissolve; **saturated** = no more solute dissolves at that temperature.
- **Solubility** of most solids **increases** with temperature; solubility of gases **decreases** with temperature.
- **Density = Mass / Volume**; SI unit = kg/m³; also g/cm³ or g/mL.
- **Relative density** = density of substance ÷ density of water (no units).
- Density **decreases** on heating (volume increases); pressure increases gas density.
- **Ice floats** because water is densest at 4°C and ice has lower density.
- Volume of irregular solids = final volume − initial volume (displacement method).
- Read the **bottom of the meniscus** for colourless liquids; top for coloured liquids.

CHAPTER 10

Light: Mirrors and Lenses

Grade 8 · Science

PROBE & PONDER

- Can we make mirrors which can give enlarged or diminished images?
- On side-view mirrors of vehicles, there is a warning: "Objects in mirror are closer than they appear." Why?
- Why is there a curved line on some reading glasses?

This chapter explores **curved mirrors** (concave and convex) and **lenses** (convex and concave), the **laws of reflection**, and the many uses of these optical devices in daily life.

10.1 What Are Spherical Mirrors?

DEFINITION

Spherical mirrors are curved mirrors shaped like a part of a hollow glass sphere. Mirrors whose reflecting surfaces are spherical are called spherical mirrors.

Type	Reflecting surface	Description
Concave mirror	Curves **inwards**	Reflecting surface curves inward (like the inner side of a spoon)
Convex mirror	Curves **outwards**	Reflecting surface curves outward (like the outer side of a spoon)

A STEP FURTHER — HOW SPHERICAL MIRRORS ARE MADE

Spherical mirrors are **not** made by slicing a hollow glass sphere. They are created by **grinding and polishing a flat glass piece** into a curved surface. If a reflective coating (thin layer of aluminium) is applied on the **outer** curved surface → **concave mirror**. If applied on the **inner** curved surface → **convex mirror**.

ACTIVITY 10.2 — DISTINGUISHING MIRRORS

Place mirrors on a table with reflecting surfaces facing upwards. View them from the side, keeping your eye at their level. The **concave mirror** has its reflecting surface curved **inwards**; the **convex mirror** has it curved **outwards**.

10.2 Characteristics of Images Formed by Spherical Mirrors

Mirror type	Object close	Object farther away
Concave mirror	Image is **erect** and **enlarged** (bigger than object)	Image becomes **inverted**; initially enlarged, then keeps getting smaller
Convex mirror	Image is always **erect** and **diminished** (smaller than object)	Image remains erect and diminished; size decreases slightly as object moves away
Plane mirror	Always **erect** and **same size** as object	Always erect and same size

- **Lateral inversion** is seen in all three types of mirrors.

10.2.1 Uses of concave mirrors

- **Reflectors of torches, headlights of cars and scooters** — concave in shape.
- **Dental mirror** — provides an enlarged view of teeth when held close to the teeth inside the mouth.
- **Reflecting telescopes** — most modern telescopes use curved mirrors, with the main mirror being a large concave mirror.

10.2.2 Uses of convex mirrors

- **Side-view mirrors on vehicles** — always form an erect image of traffic behind, smaller than actual vehicles. Since the mirror is curved outside, it provides a much **wider area** of the road behind.
- **Road safety mirrors** — installed at road intersections or sharp bends to provide drivers from both sides visibility of the other side and prevent collisions.
- **Surveillance mirrors** — installed in big stores to monitor a large area to deter thefts.

> **WHY THE WARNING ON SIDE-VIEW MIRRORS?**
> "Objects in mirror are closer than they appear" — because convex mirrors form **diminished** images, making objects look smaller and hence farther away than they actually are.

10.3 What Are the Laws of Reflection?

> **DEFINITIONS**
> **Incident ray** — the ray of light that falls on the mirror.
> **Reflected ray** — the ray of light that comes back from the mirror.
> **Normal** — the line drawn perpendicular (90°) to the mirror at the point of incidence.
> **Angle of incidence (i)** — the angle between the normal and the incident ray.
> **Angle of reflection (r)** — the angle between the normal and the reflected ray.

Laws of Reflection

1. The **angle of incidence is equal to the angle of reflection** (i = r).
2. The **incident ray, the normal at the point of incidence, and the reflected ray all lie in the same plane**.

- The laws of reflection are valid for **all kinds of mirrors** — plane, concave, and convex.
- When the incident beam falls along the normal, both the angle of incidence and angle of reflection are **zero**.

> **ACTIVITY 10.5 — REFLECTED BEAM AND PLANE**
> Shine a beam on a mirror placed on a sheet of chart paper; observe the reflected beam on the extended portion. Bend the extended portion — the reflected beam **disappears**. Flatten it again — the beam **reappears**. This shows the reflected beam lies in the same plane as the incident beam.

10.3.1 Reflection from spherical mirrors

- Even though each ray of light follows the laws of reflection, the **curved surface** of spherical mirrors causes parallel beams of rays to either **converge** (concave) or **diverge** (convex) on reflection.
- **Plane mirror** → parallel beams remain parallel after reflection.
- **Concave mirror** → parallel beams **converge** (get closer).
- **Convex mirror** → parallel beams **diverge** (spread out).
- A concave mirror is also called a **converging mirror**; a convex mirror is also called a **diverging mirror**.

> **ACTIVITY 10.7 — CONCENTRATING SUNLIGHT**
> Hold a concave mirror with its reflecting surface facing the Sun. Direct the reflected light onto a sheet of paper. Adjust the distance until you get a sharp bright spot. Hold steady for a few minutes — the paper starts to burn, producing smoke. The bright spot forms because light from the Sun gets **concentrated** at this point, producing sufficient heat to ignite the paper.

> **A STEP FURTHER — SOLAR CONCENTRATORS**
> Devices that concentrate sunlight into a small area using mirrors and lenses are called **solar concentrators**. The concentrated sunlight is used to heat a liquid to produce steam, which can generate electricity or provide heat for large-scale cooking or solar furnaces. Solar furnaces are even used for **melting steel**.

10.4 What Is a Lens?

DEFINITION

A **lens** is a piece of transparent material, usually made of glass or plastic, which has **curved surfaces**. Unlike mirrors, lenses **allow light to pass through** them, and we see things **through** a lens rather than **in** a lens.

Type	Shape	Also called
Convex lens	Thicker at the middle compared to the edges	Converging lens
Concave lens	Thicker at the edges compared to the middle	Diverging lens

10.4.1 Images formed by lenses

Lens type	Object close	Object farther away
Convex lens	Image is **erect** and **enlarged**	Image becomes **inverted**; initially enlarged, then diminishes in size
Concave lens	Image is always **erect** and **diminished**	Remains erect and diminished; size changes as distance increases

ACTIVITY 10.10 — LIGHT THROUGH LENSES

- **Thin glass plate** — light beam passes through as it is.
- **Convex lens** — light beam **converges**.
- **Concave lens** — light beam **diverges**.

ACTIVITY 10.11 — BURNING PAPER WITH A LENS

Repeat the concave mirror activity using a **convex lens** in place of the concave mirror. The convex lens converges sunlight to a sharp bright spot, which can burn paper.

Safety: Do not look at the Sun directly or through the lens — it may damage your eyes.

10.4.2 Uses of lenses

- **Eyeglasses** — lenses that help people see clearly.
- **Cameras, telescopes, microscopes** — all use lenses.
- Our **eye** has a convex lens inside it — it can change its shape, allowing us to read a book or see something far away.
- A **magnifying glass** is a convex lens that makes small print appear bigger.

OUR SCIENTIFIC HERITAGE — BHASKARA II

More than 800 years ago, during the time of the great Indian mathematician **Bhaskara II**, astronomers used **shallow bowls of water** to observe the stars and planets. By carefully looking at their reflected images through tubes placed at appropriate angles, they could measure the positions of stars and planets in the sky. Even though the laws of reflection are not mentioned in literature, their instruments and methods indicate they might have understood it in practice.

★ KEY TAKEAWAYS

- **Concave mirror** — reflecting surface curves inwards; can form enlarged, diminished, or same-size images; erect or inverted depending on distance.
- **Convex mirror** — reflecting surface curves outwards; image is always **erect and diminished**.
- **Laws of reflection:** (i) angle of incidence = angle of reflection; (ii) incident ray, normal, and reflected ray all lie in the same plane.
- **Concave mirror converges** light; **convex mirror diverges** light.
- **Convex lens converges** light; **concave lens diverges** light.
- **Convex lens** — image can be enlarged, diminished, or same size; erect or inverted depending on distance.
- **Concave lens** — image is always **erect and diminished**.
- Uses: concave mirrors in torches, headlights, dental mirrors, telescopes; convex mirrors as side-view, road safety, and surveillance mirrors.
- Uses of lenses: eyeglasses, cameras, telescopes, microscopes, magnifying glass, and the human eye.

CHAPTER 11

Keeping Time with the Skies

Grade 8 · Science

PROBE & PONDER

- Have you ever seen the Moon during the day? Why is it sometimes visible when the Sun is up?
- Imagine you lived on the Moon instead of Earth. What would you mean by a day, a month, or a year?
- What would happen if Earth had two moons instead of one?
- If we didn't have clocks or calendars, how else could we measure time?

This chapter explains the **phases of the Moon**, how **calendars** came into existence from astronomical cycles, the different types of calendars, and the role of **artificial satellites**.

11.1 How Does the Moon's Appearance Change and Why?

- The Moon is **spherical** and shines by **reflecting sunlight**.
- The Moon does not emit light of its own.
- Half of the Moon that faces the Sun receives sunlight and becomes **illuminated**; the other half remains **non-illuminated**.
- The Moon revolves around the Earth; only **one half** of the Moon always faces the Earth.
- However, the portion of the Moon facing the Earth is **not always its illuminated part**.
- We can only see the **illuminated portion** of the Moon from Earth.
- Sometimes the entire illuminated portion faces the Earth (full Moon); at other times only a part of it (crescent, gibbous, etc.).
- On New Moon day, we do not see the illuminated portion at all — only the non-illuminated portion faces the Earth.

11.1.1 Phases of the Moon

DEFINITION

The changing shapes of the bright portion of the Moon from one day to another as seen from the Earth are called the **phases of the Moon**.

- **Waning period** — the bright portion of the Moon decreases from a full circle to a half circle in about a week, and continues to shrink for another week until it is no longer visible. In India, generally called **Krishna Paksha**.
- **Waxing period** — after the new Moon, the bright side grows to a half circle in about a week and to a full circle in another week. In India, generally called **Shukla Paksha**.
- **Full Moon day (Purnima)** — the day when the Moon appears as a full bright circle.
- **New Moon day (Amavasya)** — the day when the Moon is not visible.
- The cycle from one full Moon to the next takes about a **month**.

11.1.2 Locating the Moon

- On a full Moon day, the Moon is nearly **opposite** the Sun — when the Sun rises in the East, the Moon is almost setting in the West.
- On subsequent mornings at sunrise, as its bright part decreases, the Moon appears to move **closer** in the sky to the Sun.
- When the bright part decreases to a half circle shape, the Moon is **overhead at Sunrise**.
- A few days later, the crescent Moon appears even closer to the Sun.
- A **waxing Moon** is easiest to spot at **sunset**; a **waning Moon** at **sunrise**.

- The Moon always rises and sets at **different times** than the Sun.
- The Moon rises about **50 minutes later each day**.

> **MAKING SENSE OF OBSERVATIONS**
> The shape of the Moon itself does not change — only what we see changes. The Moon revolves around the Earth; the illuminated portion of the Moon that faces the Earth changes as the Moon moves in its orbit. This change in the fraction of the illuminated portion seen from Earth causes the **phases**.

> **A STEP FURTHER — PHASES ARE NOT DUE TO EARTH'S SHADOW**
> The Moon's phases occur due to the **relative change in orientation** of the Sun, Moon, and Earth as the Moon revolves around the Earth. The Earth's shadow on the Moon causes a **lunar eclipse**, not the phases. Lunar eclipses can only happen on a **full Moon day**; solar eclipses only on a **new Moon day**. They do not occur every month because of the small tilt of the Moon's orbit with respect to Earth's orbit around the Sun.

11.2 How Did Calendars Come into Existence?

- The apparent periodic motion of the Sun — rising in the east, setting in the west — is primarily due to the **rotation of the Earth** around its own axis.
- This natural cycle of the Sun is the foundation of the **day**.
- The average time the Sun takes to go from its highest position in the sky on one day to the highest position the next day is **24 hours** — called the **mean solar day**.
- The highest position of the Sun can be found by measuring the length of shadows cast by an object — the shadow is **shortest** when the Sun is at the highest point.
- The Moon takes about **29.5 days** (nearly a month) to cycle through all its phases — the basis for a **month**.
- The Earth revolves around the Sun and takes nearly **365 and a quarter days** to complete one revolution — this cycle of seasons defines a **solar year**.

11.2.1 Lunar calendars

- In ancient times, people noticed that during one cycle of seasons, one can fit nearly **12 cycles of the phases of the Moon** (12 lunar months).
- A **lunar year** consists of 12 lunar months, totalling approximately **354 days**.
- In a lunar calendar, the **seasons do not remain synchronised** to the same lunar months in successive lunar years, because seasons repeat in approximately 365 days while the lunar year is 354 days long.

11.2.2 Solar calendars

- The need for a year to synchronise with seasons (for agriculture) led to **solar calendars**.
- The **Gregorian calendar** (widely used today) is a solar calendar.
- Months are adjusted to add up to **365 days** — some months have 30 days, others 31, and February has only 28 days.
- The Earth takes nearly an **extra quarter of a day** to complete one revolution — these extra hours add up to approximately **one day every four years**.
- To adjust, solar calendars add an extra day every four years — a **leap year**.
- In the Gregorian calendar, if a year is divisible by four, an extra leap day is added — February has 29 days.

> **A STEP FURTHER — LEAP YEAR CORRECTIONS**
> The Earth takes slightly less time than 365¼ days to go from one spring equinox to the next. Adding a day every four years helps synchronise with seasons, but adds a little too much over time. To fix this, **leap years are skipped every 100 years** — like 1700, 1800, 1900. But skipping all of them would make the calendar lag slightly behind, so every **400 years, a leap year is again added back** — like 1600 and 2000.

> **A STEP FURTHER — TROPICAL YEAR vs. SIDEREAL YEAR**

Tropical year — time between successive spring equinoxes. The Gregorian calendar is based on the tropical year.
Sidereal year — time duration required for the same stars to rise again at sunset; can also be used to define a solar calendar. The sidereal year is longer than the tropical year by a mere **20 minutes**. In modern times, astronomers use the sidereal year to keep track of Earth's position in its orbit around the Sun.

OUR SCIENTIFIC HERITAGE — UTTARAYAN AND DAKSHINAYAN

The Sun does not always rise exactly in the East. In summer, it rises a little northward of East; in winter, a little southward. These extremes happen on the **solstices**, around **June 21** and **December 21** each year.
The Sun's apparent northward movement from December to June is called **Uttarayan**; its apparent southward movement from June to December is **Dakshinayan**. This cycle repeats every year and is closely linked to the changing seasons.
The **Taittiriya Samhita** records: "Thus the Sun moves southwards for six months and northwards for six months." Ancient Indian texts like the **Surya Siddhanta** noted that the constellation **Capricorn (Makar)** would be in the background of the Sun around the winter solstice during those ancient times.

11.2.3 Luni-solar calendars

- Primarily use the Moon's phases for counting days and months but also make adjustments to stay in sync with the cycle of seasons.
- 12 lunar months add up to 354 days — fall short by nearly **11 days** compared to the solar year.
- Every 2–3 years, the accumulated difference becomes close to a full month.
- Therefore, every few years, an extra month called **Adhika Maasa (intercalary month)** is added to keep the solar year and lunar cycle in step.
- Used in many parts of India.

EVER HEARD OF... INDIAN LUNI-SOLAR MONTHS

Names of months in various Indian luni-solar calendars: **Chaitra, Vaisakha, Jyestha, Ashadha, Shravana, Bhadrapada, Ashwin, Kartika, Margashirsha (Agrahayan), Pausha, Magha, Phalguna**.
Amant — new month starts on the first day after the new Moon and ends on the day of the new Moon.
Purnimant — new month starts the day after the full Moon and ends on the full Moon.

11.2.4 The Indian National Calendar

- A national calendar by the Government of India, used along with the Gregorian calendar for multiple official purposes.
- A **solar calendar** consisting of **365 days** in a year.
- The year begins on **22 March**, the day after the spring equinox.
- Months have either 30 or 31 days. In a regular year, the **second to sixth months have 31 days** and the rest have 30 days.
- Leap years are matched to the Gregorian calendar by adding a day to **Chaitra**, the first month. In such years, the new year begins on **21 March**.
- Names of months were taken from traditional Indian calendars.

EVER HEARD OF... CALENDAR REFORM COMMITTEE

In **1952**, the Government of India set up a **Calendar Reform Committee (CRC)** to examine all existing calendars and recommend an accurate and uniform calendar. The CRC-recommended **'Unified National Calendar'** was adopted with effect from **21 March 1956 CE (1 Chaitra 1878 Saka)**. The Indian National Calendar follows the general principles of the **Surya Siddhanta**.

BE A SCIENTIST — MEGHNAD SAHA (1893–1956)

Pioneering astrophysicist of India who studied stars and their temperatures and developed a mathematical equation known as the **Saha equation**. The **Saha Institute of Nuclear Physics**, Kolkata, is named after him. He was also the **chairperson of the Calendar Reform Committee**.

11.3 Are Festivals Related to Astronomical Phenomena?

- Many Indian festivals are tied to the phases of the Moon and are based on lunar or luni-solar calendars.
- **Diwali** — new Moon of the month of Kartika.
- **Holi** — full Moon of Phalguna.
- **Buddha Purnima** — full Moon of Vaisakha.
- **Eid-ul-Fitr** — celebrated after sighting the crescent Moon at the end of the month of Ramazan.
- **Dussehra** — tenth day in the month of Ashwina.
- Festivals based on luni-solar calendars shift in the Gregorian calendar, but the shift is typically **less than a month** because the intercalary month corrects the difference.
- Purely lunar calendars do not account for this difference — festivals like Eid-ul-Fitr can occur in different months of the Gregorian calendar year after year.

> **A STEP FURTHER — SOLAR SIDEREAL FESTIVALS**
>
> A few festivals follow a **solar sidereal calendar**: **Makar Sankranti, Pongal, Bihu, Vaisakhi, Poila Baisakh, Puthandu**. These happen on almost the same date every year in the Gregorian calendar (based on the tropical year). Due to the small difference between sidereal and tropical years, their dates slowly shift — e.g. Makar Sankranti moves ahead by **one day every 71 years**. This shift is due to the slow **wobble of Earth's axis**, similar to the movement of the axis of a wobbling top.

> **EVER HEARD OF... RASHTRIYA PANCHANG**
>
> The dates of many Indian festivals are based on the exact lunar phase at sunrise. As sunrise occurs earlier in Eastern India and later in Western India, these dates can shift by a day between regions even in the same year. To maintain uniformity, the **Positional Astronomy Centre** of the Government of India annually publishes the **Rashtriya Panchang** — a detailed calculation of the positions of celestial objects for a central location in India. Based on these calculations, it provides advance intimation on dates of festivals to the Government of India for holiday declaration.

> **EVER HEARD OF... MOON IN INDIAN ART**
>
> The Moon and moonlight have inspired **ragas** in Indian classical music: **Chandrakauts, Chandranandan, Shubhapantuvarali** (means "auspicious moon"). **Mudras** in Bharatanatyam: **Chandrakala, Ardhachandran**. Traditional painting styles — **Madhubani, Warli**, and other art forms, sculpture and pottery among **Saura, Gond** and other tribes — invoke depictions of the Moon and the Sun prominently.

11.4 Why Do We Launch Artificial Satellites in Space?

- The Moon is Earth's **natural satellite**.
- **Artificial satellites** — man-made satellites sent by various countries that orbit the Earth.
- Appear as tiny specks moving in the night sky.
- Most orbit about **800 km** above Earth's surface and take roughly **100 minutes** to complete one orbit.
- Help in: **communication, navigation, weather monitoring, disaster management, scientific research**.
- **ISRO (Indian Space Research Organisation)** has launched many satellites.

> **OUR SCIENTIFIC HERITAGE — ISRO MISSIONS**
>
> - **Cartosat series** — capture high-quality images of Earth to improve maps, plan cities, handle natural disasters. **Bhuvan** — mapping platform using these images.
> - **AstroSat** — scientific observations of stars and celestial objects.
> - **Chandrayaan 1, 2, 3** — to the Moon.
> - **Aditya L1** — to study the Sun.
> - **Mangalyaan** — to Mars.
> - Student satellites: **AzaadiSat, InspireSat-1, Jugnu**.

A STEP FURTHER — SPACE DEBRIS

After their useful life, many artificial satellites and their rocket parts become **space junk or space debris**. This debris crowds space and could collide with working satellites. Small debris burns up in the atmosphere when it falls towards Earth; larger pieces can crash on the ground. Countries are now working together to remove this dangerous debris.

BE A SCIENTIST — VIKRAM AMBALAL SARABHAI (1919–1971)

Researcher in space science and nuclear physics; known as the **Father of the Indian Space Programme**. Pioneered the effort to launch the first artificial satellites. The **Vikram Sarabhai Space Centre (VSSC)**, Thiruvananthapuram — the ISRO centre that develops rockets and launch vehicle technology — is named after him.

★ KEY TAKEAWAYS

- **Phases of the Moon** occur because we see different parts of the illuminated portion as the Moon revolves around Earth.
- Full cycle of phases ≈ **29.5 days (about a month)**.
- **Waxing** (Shukla Paksha) — bright part increases; **Waning** (Krishna Paksha) — bright part decreases.
- **Full Moon (Purnima)** = full bright circle; **New Moon (Amavasya)** = not visible.
- **Lunar calendars** follow the Moon's cycle (354 days); **solar calendars** follow seasons (365 days); **luni-solar calendars** combine both with an intercalary month (Adhika Maasa).
- **Gregorian calendar** is a solar calendar; leap year adds Feb 29 every 4 years.
- **Indian National Calendar** — solar; year begins 22 March; 365 days; leap years add a day to Chaitra.
- Many Indian festivals are tied to astronomical phenomena — Diwali (new Moon), Holi (full Moon), etc.
- **Artificial satellites** help in communication, navigation, weather monitoring, disaster management, scientific research.

CHAPTER 12

How Nature Works in Harmony

Grade 8 · Science

PROBE & PONDER

- How might loss of forest cover and changes in rainfall patterns lead elephants to enter human farms and villages?
- Imagine you are a tree in a dense forest. What kind of relationships would you have with water, sunlight, other animals, and other components of the forest?
- Do you think the Earth can thrive without humans? Can humans survive without the Earth?
- If two kinds of birds compete for the same fruit, how might their way of living change over time?
- Can human actions cause natural disasters?

This chapter explores **habitats**, **populations**, **communities**, **ecosystems**, **food chains and food webs**, **decomposers**, **interactions among organisms**, and the **benefits of ecosystems** — ending with threats like deforestation and the need for conservation.

12.1 How Do We Experience and Interpret Our Surroundings?

DEFINITION

A **habitat** is a place where an organism lives. It could even be just the bark of a tree.

- Different habitats have different kinds of plants and animals.
- The plants and animals interact with each other and adapt to survive in the surrounding conditions.
- Every organism needs **specific conditions** to survive.
- Different habitats offer different living conditions.
- A pond provides food, oxygen, shelter, and space — conditions essential for survival of organisms.
- Fish obtain biotic needs (food) from small plants and animals, and abiotic needs (oxygen) from water.

Component	Definition	Examples
Biotic components	Living beings in a habitat	Fish, frogs, turtles, snakes, dragonflies, mosquitoes, snails, ducks, algae, diatoms, duckweeds, lotus, grass, trees, birds
Abiotic components	Non-living things in a habitat	Air, sunlight, water, temperature, soil

- Different organisms living in the same habitat may use the resources in **different ways**.
- A snake comes out at night; a rodent is active during the day — both live in the same habitat but face different conditions.
- This is how living organisms **coexist in harmony** in the same habitat.

12.2 Who All Live Together in Nature?

DEFINITION

A **population** is a group of the same type of organisms in a habitat at a given time.

- We can observe and record populations of different kinds of organisms in a single habitat.
- If all organisms were the same, they would have the same requirements — food, water, space — leading to **competition** and possible scarcity of resources.

DEFINITION

> A **community** comprises different populations sharing the same habitat. The biotic components of a habitat — plants, animals, and microorganisms — together form the community.

- These organisms interact and depend on one another for survival.

> **EVER HEARD OF... POLLINATION**
> A flower has a stalk, green leafy structures called **sepals**, coloured **petals**, and two reproductive parts — **carpels** (female) and **stamens** (male). Stamens burst and release yellow dust-like **pollen grains**. Wind, water, insects, bats, and birds help carry pollen from stamens to carpels of the same and/or different flowers. This process is called **pollination** — essential for the formation of fruits and seeds.

12.3 Does Every Organism in a Community Matter?

- Study comparing two ponds: **Pond A with fish** and large number of flowering plants; **Pond B without fish** and fewer flowering plants.
- **Observation:** Pond A (with fish) had **fewer dragonflies** than Pond B.
- **Reason:** Fish eat dragonfly larvae, so ponds with fish had fewer dragonflies.
- Dragonflies usually eat flies, bees, and butterflies. With fewer dragonflies, **more bees, flies, and butterflies** were found.
- These insects help **pollinate flowers** from nearby areas, moving pollen from one flower to another, which helps plants produce **seeds**.
- So, flowers near ponds with fish may produce **more seeds** than those near ponds without fish.

> **KEY INSIGHT**
> This study shows how biotic components (fish, dragonflies, pollinators, plants) and abiotic components (temperature, water, nutrients) **interact with and affect each other**. Overfishing by humans can change this balance and affect the living and non-living parts of the habitat.

12.4 Interactions Among Organisms and Their Surroundings

- Living organisms (biotic community) depend on non-living things (abiotic components) for survival.
- Plants and animals also depend on each other for nutrition, respiration, and reproduction.
- Both types of interactions — among biotic components, and between biotic and abiotic components — are important for survival in any habitat.

Criterion	Description	Examples
Interaction between **abiotic and biotic**	Influence life processes like nutrition, respiration, reproduction	Earthworms live in moist soil; many microbes in the pond; fish lays eggs in water
Interaction between **two abiotic**	Influence physical characteristics of a habitat	Day temperature is high due to bright sunlight; water evaporating due to sunlight; air current creating gentle waves
Interaction among **biotic components**	Influence availability of resources needed for life processes	Frog eats insects; water snake eats fish; frogs and fish may compete for small insect larvae; fish lays eggs near vegetation to protect from other fish or frogs

> **DEFINITION**
> An **ecosystem** is formed by the interaction between biotic components (plants, animals, microorganisms) and abiotic components (air, water, soil, sunlight, temperature) in a habitat.

- **Aquatic ecosystems** — ponds, rivers, lakes.
- **Terrestrial ecosystems** — forests, farms, large trees like banyan, mango, pilkhan.
- Ecosystems can be **large or small**.

- Ecosystems can **overlap** — e.g. a small river (aquatic) along with mountains, forests, grassland, farmland (terrestrial).
- **Farmland** is a human-made ecosystem.
- Abiotic components are also influenced by biotic components — plants release oxygen during photosynthesis, roots hold soil in place and prevent erosion, plants retain soil moisture and help cool the atmosphere.

12.5 Who Eats Whom?

DEFINITIONS

Producers (autotrophs) — organisms that make their own food by photosynthesis (auto = self, troph = food). Green plants.
Consumers (heterotrophs) — organisms that cannot produce their own food and depend on other organisms for food (hetero = other, troph = food).
Herbivores — eat only plants (deer, hare).
Carnivores — eat only animals (leopard, snake, eagle).
Omnivores — eat both plants and animals (crows, foxes, mice).
Decomposers (saprotrophs) — break down dead matter and recycle nutrients (bacteria, fungi; sapro = rotten, trophs = food).

12.5.1 Food chain

DEFINITION

A **food chain** is a simple sequence showing 'who eats whom' in an ecosystem. The interactions between biotic components based on feeding relationships can be represented as a linear chain.

- Example (grassland): **Grass → Grasshopper → Frog → Snake → Eagle**
- Example (crop field): **Millet → Mouse → Eagle**

12.5.2 Trophic levels

- Each organism in a food chain has a specific position called a **trophic level** (troph = food).
- **Producers** (green plants) — first trophic level.
- **Herbivores** (hares, deer) — second level.
- **Small carnivores** (frogs) — third level.
- **Large carnivores** (tigers, vultures) — next level.

12.5.3 Food web

- Each organism may be eaten by **two or more** types of organisms.
- In an ecosystem, food chains are **interlinked** with each other to form a network called a **food web**.

12.6 What Happens to Waste in Nature?

- Organisms produce a lot of waste, including dead matter and food waste.
- Mushrooms (a type of fungi) grow on dead matter.
- Microorganisms like **fungi and bacteria** break down complex substances in dead plants and animals into simpler ones.
- This process returns important nutrients to the soil.
- Tiny insects like beetles and flies on animal droppings (e.g. elephant dung) help break it down and recycle nutrients.
- This process is called **decomposition**; the organisms carrying it out are **decomposers (saprotrophs)**.
- Plants grow in soil; many nutrients in soil come from decomposition. Decomposers play an important role in **recycling nutrients**.

- In nature, **nothing is wasted** — everything is reused.

> **EVER HEARD OF... MIGRATORY BIRDS**
>
> Many migratory birds fly thousands of miles and reach different habitats in India. They migrate to avoid harsh climate and in search of food. Birds play a significant role in keeping the balance in an ecosystem as **pollinators or seed dispersers** along the path they migrate, linking two habitats. They are predators of insect pests and help farmers control pest populations. Example: **Demoiselle Crane** visits the water body of **Khichan village, Jodhpur district** during winter.

12.7 How Does One Change Lead to Another?

- Example chain: Plants in a pond die → less oxygen produced in water → fish population drops → fewer consumers in the pond → insects increase in number → insects spread to nearby farmlands → farmers use pesticides → pesticides adversely affect the environment → further environmental issues.

> **ACTIVITY 12.9 — FROG LEGS CASE STUDY**
>
> In the 1980s, India was a significant exporter of frog legs, especially of the **Indian bullfrog** (*Hoplobatrachus tigerinus*). Large-scale harvesting led to a decline in frog populations. Since frogs eat insects, their reduced numbers resulted in a **rise in agricultural pests**. This forced farmers to use more synthetic pesticides, which harmed the environment, soil, and water quality, and affected environmental and human health. The Government of India **banned the export of frog legs** to prevent further ecological damage.

12.8 How Do Interactions Maintain Balance in Ecosystems?

- Besides feeding relationships, organisms also **compete** for common resources like food, water, physical space, or sunlight.
- This competition helps control population size and keeps the ecosystem balanced.
- Without it, one species could multiply too much, causing an imbalance.
- An ecosystem stays in balance when interactions among organisms and their environment keep populations and resources stable.
- This balance is **dynamic**, not fixed, and can be disrupted by natural or human-made changes.

Interaction	Description	Example
Mutualism	Both organisms benefit	Honeybees and flowers — flower gets pollinated, bee gets nectar
Commensalism	One organism benefits, the other is not affected	Orchids on trees — orchid gets physical support, tree branch is unaffected
Parasitism	One organism benefits, the other is harmed	Ticks on the body of dogs — tick feeds on dog blood (benefited), dog gets skin irritation (harmed)

> **BE A SCIENTIST — ASIR JAWAHAR THOMAS JOHNSINGH (A.J.T. JOHNSINGH)**
>
> Famous Indian wildlife biologist who helped us understand forest ecosystems through the eyes of animals. A pioneer in studying wildlife through modern tracking systems. His research showed how predators like tigers and leopards rely on prey such as deer and wild boar (working in **Bandipur National Park, Karnataka**). He proved that a **healthy prey population is key to predator survival** and inspired many youngsters to study wildlife and protect forests.

12.9 What Are the Benefits of an Ecosystem?

- Biotic and abiotic components of an ecosystem depend on each other and support various life processes.
- Humans also benefit from ecosystems.
- **Forests** provide fresh air, fertile soil, food, fibres, timber, and medicines.
- **Aquatic ecosystems** provide water and food.

- Ecosystems also offer **aesthetic and recreational value**.
- However, when we **overuse or misuse** natural resources, we disturb the balance in nature.

12.9.1 Case study — The Sundarbans

- The **Sundarbans** have the largest **mangrove forests** in the world.
- Located where the **Ganges and Brahmaputra Rivers** meet between India and Bangladesh.
- Home to various flora and fauna, many of which are endangered.
- The Sundarbans protect us by **slowing down strong winds and waves** during storms and floods.
- Trees absorb carbon dioxide from the air and release oxygen.
- UNESCO declared the Sundarbans a **World Heritage Site in 1987**.
- **Threats:** Mangrove trees being cut for fuelwood and farming; illegal hunting and overuse of forest resources; pollution from industrial waste and untreated sewage in rivers.

12.9.2 Human-made ecosystems

- Humans have created artificial ecosystems like **fish ponds, farms, and parks** to meet their needs.
- When well designed, these can help **reduce pollution, support biodiversity, and provide recreational spaces**.
- Unlike natural ecosystems, these need **human care and management**.

12.9.3 How do healthy ecosystems serve our farms?

- Farming is a major livelihood in India; can become unsustainable if not managed well.
- Between **1950 and 1965**, India faced a food crisis due to low crop production.
- In the mid-20th century, the use of tractors, machines, synthetic fertilisers, and pesticides helped increase food production — this period is known as the **Green Revolution**.
- However, these methods are now considered **unsustainable** because of overuse of synthetic chemicals, excessive groundwater extraction, and growing only one type of crop for commercial gain.
- **Effects of overuse:** Soil degradation; reduced soil fertility (decrease in friendly microorganisms and organic matter/humus); soil prone to erosion; reduction in natural predators → increase in pests; some pests develop **resistance** to pesticides.
- **Monoculture** (growing the same crop repeatedly) reduces biodiversity and affects pollinators, crucial for food production.
- **Organic and natural farming** methods aim to reduce use of synthetic fertilisers and support sustainable farming with minimal interference in natural ecosystems.

> **OUR SCIENTIFIC HERITAGE — VRIKSHAYURVEDA**
> The ancient text **Vrikshayurveda** emphasises soil health and nourishment. It strongly advocates for the continuous nourishment of the soil through organic manure like **Kunapa Jala** (a liquid fertiliser made from animal and plant waste by fermentation) and other composted materials.

★ KEY TAKEAWAYS

- **Habitat** = place where an organism lives; has **biotic** (living) and **abiotic** (non-living) components.
- **Population** = same type of organisms; **community** = different populations sharing a habitat.
- **Ecosystem** = interaction between biotic and abiotic components; can be terrestrial or aquatic; can overlap.
- **Producers** (plants) → **consumers** (herbivores, carnivores, omnivores) → **decomposers** (bacteria, fungi).
- **Food chain** = who eats whom; **food web** = interconnected food chains; **trophic levels** = positions in a food chain.
- **Decomposers** break down dead matter and recycle nutrients — essential for soil fertility.
- Interactions: **mutualism** (both benefit), **commensalism** (one benefits, other unaffected), **parasitism** (one benefits, other harmed).
- Ecosystems provide clean air, water, food, medicine, and climate regulation; threatened by pollution, deforestation, invasive species, overexploitation.
- **Protected areas** (national parks, sanctuaries, biosphere reserves) help conserve wildlife.

CHAPTER 13

Our Home: Earth, a Unique Life Sustaining Planet

Grade 8 · Science

PROBE & PONDER

- What do you think Earth would look like if there were no life on it at all?
- Life on Earth has survived for billions of years. What allows it to keep going despite major changes and disasters?
- Why don't dogs lay eggs? Or hens give birth to live chicks?
- If a spaceship carried soil and water to Mars, could plants start growing there?

This final chapter explains why Earth is uniquely suited for life — its **distance from the Sun**, **size**, **atmosphere**, **magnetic field**, and the delicate **balance** of its systems — then covers **reproduction** as the key to continuity of life, and finally the **threats** facing life on Earth.

13.1 Why Is Earth a Unique Planet?

- Earth is the only planet where life, as we know it today, exists and thrives in all its forms.
- All life on Earth exists in a very thin layer on the surface of our planet.
- From the tallest mountain to the deepest ocean trench, the **crust** where all life exists is tiny compared to the size of Earth.
- If Earth were the size of an apple, the crust would be as thin as the apple's skin.
- This delicate, life-supporting layer is what makes Earth truly special.

13.2 What Do the Planets of Our Solar System Look Like?

- Our solar system has **eight planets** that go around the Sun in nearly circular orbits.
- In order of increasing distance from the Sun: **Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune**.
- **Mercury, Venus, Earth, Mars** — relatively small and **rocky** planets.
- **Jupiter, Saturn, Uranus, Neptune** — large planets, mostly made of **gases**.

Planet	Average temperature	Radius (Earth = 1)	Atmosphere?
Mercury	170°C	0.38	No
Venus	450°C	0.95	Yes
Earth	15°C	1	Yes
Mars	−60°C	0.53	Yes (very thin)
Jupiter	−110°C	11.2	Yes
Saturn	−140°C	9.4	Yes
Uranus	−195°C	4.0	Yes
Neptune	−200°C	3.9	Yes

WHY IS VENUS THE HOTTEST PLANET?

Venus is the hottest planet not because it is the closest to the Sun, but because its **thick atmosphere traps heat**. The air on Venus is almost entirely made up of **carbon dioxide**, which does not let heat escape. This is called the **greenhouse effect** — it makes Venus even hotter than Mercury, which is closer to the Sun.

A STEP FURTHER — GREENHOUSE EFFECT vs. GREENHOUSE

> The greenhouse effect that causes a planet like Venus and Earth to trap heat does **not** work the same way as a greenhouse for growing plants. On Venus or Earth, gases like carbon dioxide in the atmosphere trap heat by absorbing the radiation given off by the Earth after it gets warmed by the Sun. A plant greenhouse, on the other hand, traps warmed air simply because it is a **closed space**, usually with glass walls. Both keep things warm, but they do it differently.

13.3 What Makes Earth Suitable for Life?

13.3.1 Distance from the Sun — The Habitable Zone

- Earth is at the **right distance** from the Sun where temperature allows water to exist in **liquid form**.
- If Earth were closer to the Sun, it would be too hot and all water would evaporate.
- If Earth were farther away, it would be too cold and all water would freeze.
- Liquid water is essential for life to evolve (from what we know so far).
- The range of distances from the Sun (or another star) over which water remains liquid is called the **habitable zone** or **'Goldilocks zone'**.
- Most of Earth's surface is covered with water — hence the name **Blue Planet**.

> **A STEP FURTHER — DID MARS EVER SUPPORT LIFE?**
> Mars lies at the **edge of the Sun's habitable zone**. Several spacecraft have been sent to Mars; rovers have landed and explored its surface, but **no proof of life has been found yet**. However, scientists think that in the past, Mars may have had **liquid water** — maybe even some lakes — and conditions that could support simple life forms. This is one reason Mars continues to interest scientists.

13.3.2 Size of the Earth

- In our solar system, the orbits of most planets including Earth are almost **circular** — this keeps the amount of sunlight and heat nearly steady throughout the year, preventing extreme summers and winters at most places.
- Earth is the **right size** to support an atmosphere.
- If Earth were much smaller (with the same average density), its **gravity would be too weak** to hold on to the gases in our atmosphere — they would escape into space.
- Mars has an atmosphere **100 times thinner** than Earth; Mercury has **no atmosphere** at all.
- If Earth were too large and gravity much stronger, it would pull us down with such a large force that our **bones could get crushed**.
- Due to the right size, Earth can support an atmosphere essential for life.
- **Oxygen** in the atmosphere allows us to breathe and is needed by almost all forms of life.
- Some oxygen gets converted to **ozone** (a three-atom oxygen molecule), forming the **ozone layer** — acts like a shield, blocking harmful **ultraviolet (UV) rays** from the Sun that can damage living cells.

13.3.3 Magnetic Field of the Earth

- Earth itself behaves like a giant magnet.
- It is believed that the movement of **molten iron in Earth's core** may be the origin for Earth's magnetic field.
- Earth is constantly hit by tiny, high-energy particles from space: **cosmic rays** (from far across the universe) and **solar wind** (from the Sun).
- These particles can damage the atmosphere, reduce the ozone layer, and let in more harmful UV rays.
- Earth's magnetic field acts like a **protective shield** — pushes many harmful particles away, keeping our atmosphere and life safe.

> **OUR SCIENTIFIC HERITAGE — MANGALYAAN (MARS ORBITER MISSION)**
> Launched in **2013** by ISRO — a big step in exploring Mars. Carried tools to study the planet's atmosphere, surface, and signs of past water. Showed the world that India could do space science with smart, low-cost technology.

13.4 What Allows Life to Be Sustained on Earth?

13.4.1 Air, water, and sunlight

- Atmosphere contains **oxygen** for respiration (humans, animals, plants).
- In the presence of **sunlight**, plants take carbon dioxide from air and water from soil to prepare food by **photosynthesis** — releasing oxygen.
- Radiation from the Sun heats the Earth; some heat is trapped by the atmosphere due to the **greenhouse effect** — keeps temperature just high enough for water to remain liquid.
- Without an atmosphere, Earth would lose heat to space and become too cold.
- **Water** covers about **70%** of Earth's surface — found in ponds, lakes, rivers, springs, seas, oceans, and groundwater. All this water forms the **hydrosphere**.
- Water is a good solvent; transports nutrients from soil to leaves in plants; in animals, regulates body temperature, aids digestion, ensures hydration.
- Hydrosphere is home to millions of life forms — from tiny plankton to giant whales.
- Water vapour in air forms clouds and brings rain or snow — refills rivers, lakes, and underground water.
- Rainfall affects the kinds of plants and animals that can live in a place.
- Moving air shapes weather and rainfall — influencing farming, water supply, and life on land.

13.4.2 Soil, rocks, and minerals

- Earth's crust is made of **rocks, soil, and minerals**.
- Provides almost everything life needs to grow and survive — from soil for plants to minerals giving us salt, coal, oil, and metals like iron and copper.
- The solid parts of Earth, including rocks, soils, and minerals, are known as the **geosphere**.
- Soil is rich in nutrients like **nitrogen and potassium** that plants need — come from slow breakdown of rocks and remains of plants and animals.
- **Geodiversity** — the variety of landforms, rocks, soils, etc., along with the processes that shape and alter them. Helps create unique habitats where different types of life can thrive.

13.4.3 Plants, animals, and microorganisms

- Earth is full of life — trees, shrubs, herbs, animals, insects, and tiny organisms invisible to the naked eye.
- All living beings, along with the places where they live, make up the **biosphere**.
- Includes land, water, and air, where life interacts with its surroundings to survive and grow.
- Plants make food through photosynthesis; animals eat plants or other animals; decomposers break down dead matter and return nutrients to the soil.
- Nature works together as a **system** to support life.

13.4.4 The importance of balance

- Earth is like a giant teamwork project between nature, weather, and life itself.
- A vast, living system where land, air, water, and living things support and affect one another.
- Even a small change in one part — like cutting down a forest — can impact rainfall, soil, air quality, and the animals that live there.
- Life on Earth survives because everything works together in **balance**.
- Preserving and protecting clean air, water, soil, and all forms of life is essential for keeping Earth healthy for the future.

13.5 What Keeps Life from Disappearing?

- If plants and animals didn't reproduce, life would eventually disappear from Earth.
- **Reproduction** ensures each type of organism continues to exist, maintaining the continuity of life.

- Parents pass on instructions to their offspring about how to develop from a single cell — called **genetic material** or **genes**, stored inside every cell.
- Genes = a detailed instruction manual inside each cell — some instructions tell the cell how to make blood, others guide formation of bones, muscles, or skin.
- Reproduction also allows for **small changes** in the instructions passed down — sometimes helping a plant or animal survive better in a new environment.
- Example: Over time, camels developed humps to store fat and survive in deserts. Some bacteria have become resistant to antibiotics.
- Over many generations, such changes can lead to new features — or even completely new types of living beings.
- Reproduction not only keeps each kind continuing but also helps life **adapt, change, and continue in new forms**.

13.5.1 Asexual reproduction

> **DEFINITION**
> In **asexual reproduction**, a single parent produces new individuals that are **exact copies** of itself (exact in terms of the instructions inside the cell).

- Many plants can reproduce when any part — leaf, stem, or root — is planted in soil. This is called **vegetative propagation**.
- Examples: money plant (stem cutting), potato ('eyes'), ginger.
- Single-celled organisms like **bacteria and amoeba** divide into two identical individuals.
- Some multicellular organisms like **algae** can regrow from small cut parts.
- **Hydra** grows tiny buds on its body that break off and grow into new individuals.
- **Planaria** (a type of flatworm) can regrow from a fragment of its body — studied by scientists to understand regeneration.

13.5.2 Sexual reproduction

> **DEFINITION**
> In **sexual reproduction**, two parents are involved — usually called **male and female**. Instructions from two parents combine to create offspring that are **not exactly like either parent**.

- Flowering plants have male and female parts.
- Some microorganisms like bacteria and yeast also have two 'mating types' that act like the two parents.
- Each parent makes specialised reproductive cells called **gametes** — these carry only **half** of the parent's genetic material.
- When male and female gametes join, they form a new cell with a complete set of instructions — half from each parent.
- Each gamete carries a different set of instructions for things like eye colour, hair type, and more.
- These instructions mix in different ways each time, so each child is **unique** — one child may inherit a nose similar to the mother, another may inherit eyes similar to the father.

Sexual reproduction in plants:

- **Pollen grains** found inside the **anther** of a flower are the male gametes.
- **Ovules**, found deep inside the flower, are the female gametes.
- Pollen is carried to another flower by wind, insects, or animals — this process is called **pollination**.
- When male and female gametes combine, it is called **fertilisation**, forming a **zygote** that becomes the seed.
- The fleshy part of the flower around the ovule develops into a **fruit**.

- When birds or animals eat the fruit, the seeds often get dropped far from the original plant — a helpful way for plants to spread.
- When seeds get water, they use stored nutrients to grow roots and shoots — **germination**.

Sexual reproduction in animals:

- Gametes are called **sperm** (male) and **eggs** (female).
- In fish and frogs, fertilisation takes place in **water** — male and female eject sperm and eggs into the water where they combine to form the zygote. Development of the zygote into an embryo also takes place in water.
- In **birds and mammals** (including humans), sperm are deposited inside the body of a female and fertilisation takes place when the sperm swim towards the egg.
- **Birds**: the fertilised egg (zygote) is 'laid' by the female. Development of the zygote into an embryo happens after the egg is laid, during the **hatching** process. The female parent puts a lot of 'food' into each egg — has to last for the developing embryo until it hatches.
- **Most mammals**: development of the zygote into an embryo takes place **inside the body** of the female. The mother's body provides all the food and oxygen the baby needs to grow until it is born.

13.6 What Are the Threats to Life on Earth?

- Human actions are disturbing the delicate balance of living and non-living things.
- Even small changes in global temperature, oxygen levels, or the ozone layer can put life at risk.
- The biggest environmental challenges we face today are **climate change, biodiversity loss, and pollution** — together known as the **triple planetary crisis**.

13.6.1 Climate change

- Burning **fossil fuels** like coal and oil releases **greenhouse gases** like carbon dioxide and methane.
- These trap even more heat in the atmosphere, causing **global warming**.
- Normally, Earth keeps a balance — trees, plants, and tiny ocean plankton absorb carbon dioxide as they grow.
- But when we burn fossil fuels, we release extra carbon that has been locked underground for millions of years. The Earth cannot absorb this fast enough, so heat builds up.
- Even a small increase in temperature can melt ice caps, raise sea levels (flooding coastal cities), cause extreme weather conditions, and lead to many plants and animals disappearing.
- These long-term changes in temperature, rainfall, and weather patterns are called **climate change**.

13.6.2 Biodiversity loss

- When natural habitats are destroyed, plants and animals may disappear, upsetting ecosystems.
- Example: If grasses vanish, animals that feed on them (deer, grasshoppers) struggle to survive. Without herbivores, predators like tigers or foxes lose their food too.
- Every type of living thing has a role; losing even a few weakens nature's ability to support life.

13.6.3 Pollution

- Air pollution from factories, vehicles, and burning fuels harms both people and nature.
- Can cause breathing problems, damage crops, and lead to **smog and acid rain**.
- Water and soil pollution are serious threats to life.
- Factory, farm, and plastic waste harm aquatic life and make water unsafe.
- Excess fertilisers and poor waste disposal pollute soil, reduce crop yield, and spread harmful substances through the food chain.

> **EVER HEARD OF... GLOBAL AGREEMENTS**
>
> - **Montreal Protocol (1987)** — helped reduce harmful chemicals like Chlorofluorocarbons (CFCs), allowing the ozone layer to slowly recover.
> - **Earth Summit (1992)** — led to international efforts on climate change and biodiversity.

- **Kyoto Protocol (2005)** and **Paris Agreement (2015)** — committed countries to reduce greenhouse gas emissions.
- **Paris Agreement** set a goal to limit global warming to below **1.5°C**, but as of 2025, the world is not on track to meet that goal.

13.6.4 Protecting life on Earth — what can be done

- Cut down on greenhouse gases by using **renewable energy** like solar and wind.
- Improve energy use and choose environmentally friendly ways to travel.
- Preserve **biodiversity** — diverse ecosystems are stronger and more balanced.
- Local communities can play a big role in using natural resources sustainably.
- Reuse, repair, and recycle items like clothes and plastic to reduce pollution and waste.
- Small actions like saving energy and water add up.
- Learn more, share ideas, and encourage others.

★ KEY TAKEAWAYS

- Earth is unique because of its **distance from the Sun** (habitable zone / Goldilocks zone), its **size**, its **atmosphere**, and its **magnetic field**.
- Liquid water is essential for life; Earth's distance allows water to remain liquid.
- Earth's **magnetic field** shields it from harmful cosmic rays and solar wind.
- **Ozone layer** blocks harmful UV rays.
- Earth's systems: **atmosphere** (air), **hydrosphere** (water), **geosphere** (solid Earth), **biosphere** (life) — all interact to sustain life.
- **Reproduction** ensures continuity of life; **asexual** (exact copies) vs. **sexual** (offspring different from parents).
- In sexual reproduction, **gametes** carry half the genetic material from each parent; fertilisation forms a **zygote**.
- **Triple planetary crisis** = climate change + biodiversity loss + pollution.
- Global agreements: Montreal Protocol (1987), Earth Summit (1992), Kyoto Protocol (2005), Paris Agreement (2015).